Scientific Reports

https://doi.org/10.1038/s41598-026-37984-z

Article in Press

Anti interference and fault tolerant control of UAVs integrating residual based diagnosis disturbance estimation with counter drone strategies

Received: 22 October 2025

Accepted: 28 January 2026

Published online: 17 February 2026

Cite this article as: Xie Z. & Long Y. Anti interference and fault tolerant control of UAVs integrating residual based diagnosis disturbance estimation with counter drone strategies. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-37984-z

Zheyi Xie & Yucai Long

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

Anti Interference and Fault Tolerant Control of UAVs Integrating Residual Based Diagnosis Disturbance Estimation with Counter Drone Strategies

Zheyi Xie[1]*, and Yucai Long [1]

1. Mechatronics and Control Engineering School of Loudi Vocational and Technical College Loudi 417000, Hunan, China, lunmia78005@smc.edu.kg, bran64632@smc.edu.kg

* Correspondence: lunmia78005@smc.edu.kg, zheyixieedu@163.com

Abstract

Unmanned Aerial Vehicles (UAVs) are increasingly deployed in complex, uncertain, and adversarial environments, yet they remain vulnerable to actuator faults, environmental disturbances, and deliberate interferences such as jamming or spoofing. Conventional control approaches typically address these challenges independently, focusing on either fault-tolerant control (FTC), disturbance rejection, or counter-UAV defense. This paper presents an integrated anti-interference and fault-tolerant control framework that unifies three complementary modules: (i) residual-based fault detection and isolation, (ii) adaptive extended state observer (AESO)-based disturbance estimation and compensation, and (iii) counter-UAV evasive maneuver strategies. The framework is formulated using a nonlinear UAV dynamic model and rigorously analyzed under Lyapunov stability theory. Simulation results demonstrate significant performance improvements compared with baseline controllers. The proposed method reduced position and attitude deviations to below 0.05 m and 0.03 deg, respectively, under simultaneous actuator faults and wind disturbances. Fault and disturbance estimation errors remained below 0.05 and 0.03 units, respectively, ensuring timely control reconfiguration. Furthermore, the UAV achieved a 100% success rate in counter-drone evasive maneuvers while maintaining trajectory stability. These results confirm that the integrated design provides high resilience, rapid recovery, and reliable performance in fault-disturbed and adversarial conditions. By bridging FTC, adaptive disturbance rejection, and counter-UAV defense, the proposed framework advances the state of the art in resilient UAV control for civilian and defense applications.

Keywords: UAVs; fault-tolerant control; anti-jamming; fault detection and isolation (FDI); disturbance estimation; counter-UAVs defense; resilience

ARTICLE IN PRESS

ARTICLE IN PRESS

1. Introduction

The rapid evolution of autonomous aerial systems has redefined the landscape of modern aviation, where resilience, adaptability, and intelligent control are now paramount to mission success. As UAVs transition from controlled laboratory environments to real-world, dynamic, and adversarial conditions, their exposure to unpredictable disturbances, sensor and actuator degradations, and cyber-physical interferences has intensified. Conventional control strategies, typically designed for nominal conditions, often struggle to maintain stability and performance when confronted with simultaneous system faults and environmental uncertainties. This growing operational complexity necessitates integrated, process-oriented control frameworks capable of detecting faults, compensating disturbances, and countering external interferences in real time. Against this backdrop, the present study introduces a unified anti-interference and fault-tolerant control approach that bridges theoretical advancements in adaptive estimation and robust control with practical demands of resilient UAV operation.

Unmanned Aerial Vehicles (UAVs) have transitioned from niche experimental platforms into indispensable tools across diverse applications, including environmental monitoring, logistics, defense surveillance, and disaster response [1-4]. Their versatility, low operational cost, and capability to perform missions in hazardous environments make them attractive alternatives to manned aircraft. However, as UAVs grow more autonomous and operate in increasingly complex airspaces, ensuring their safety and reliability has become a pressing concern [5-8]. A primary challenge is that UAVs are inherently vulnerable to systemic faults, environmental disturbances, and deliberate adversarial interferences. Addressing these threats requires a comprehensive control strategy that integrates fault-tolerant mechanisms with interference mitigation [6-8].

First, internal system faults in UAVs can manifest in numerous forms, such as actuator degradation, control surface jamming, or sensor bias and drift. For example, partial thrust loss in a quadrotor's motor can destabilize flight, while erroneous gyroscope measurements can mislead the controller into executing erroneous commands [9-12]. Recent studies have increasingly demonstrated that advanced control schemes particularly those employing disturbance observers, adaptive mechanisms, and evolutionary optimization, offer substantial benefits in handling uncertainties, external disturbances, and real-time operational constraints. For example, disturbance-observer-based and adaptive gain-learning approaches have shown significant improvements in robustness and transient performance, as evidenced by differential-evolution-driven high-gain tuning strategies that accelerate controller adaptation in nonlinear environments [13], as well as observer-based differential evolution constrained controllers designed to achieve safe tracking in robotic systems under bounded inputs [14]. Likewise, observer-based event-triggered frameworks have been proposed to enhance networked control resilience against attacks and communication imperfections, demonstrating the growing integration of intelligent estimation techniques within advanced control architectures [15].

ARTICLE IN PRESS

In parallel, meta-heuristic and fuzzy-adaptive methodologies have also contributed to broadening the capabilities of modern control systems. Bat-algorithm-based controllers, for instance, have been utilized to reduce energy consumption and improve trajectory accuracy in robotic manipulators, highlighting the effectiveness of biologically inspired optimization in dynamic environments [16]. Furthermore, adaptive decentralized finite-time fuzzy controllers have been developed to ensure secure operation of cyber–physical systems under deception attacks, offering enhanced resilience and strong stabilization properties in the presence of nonlinear uncertainties [17]. Additional work on hierarchical and multi-stage controllers, for example, a three-tiered structure designed for obstacle avoidance in PV-powered mobile robots, illustrates how adaptive and observer-enhanced schemes can be embedded into complex electromechanical systems to achieve reliable operation under practical constraints [18]. These contributions collectively underscore that the integration of disturbance-observer-based and adaptive methods is broader than previously stated, and they motivate a more unified treatment of fault detection and advanced control within the present study.

Conventional Fault-Tolerant Control (FTC) systems aim to preserve stability and acceptable performance despite such failures. Passive FTC designs rely on redundancy, while active designs incorporate fault detection and reconfiguration. Yet, existing methods often assume benign operating conditions and rarely consider the compounded effect of faults and adversarial interference [10-12, 18, 19]. Second, external disturbances such as wind gusts, turbulence, and aerodynamic uncertainties are pervasive in UAV operations. The effects of these disturbances are amplified for small-scale UAVs due to their low inertia and sensitivity to environmental conditions. Classical control techniques like PID or LQR often fail to ensure robustness in highly nonlinear conditions. Advanced control schemes, such as disturbance observers and adaptive controllers, provide improved rejection but lack integration with fault detection schemes [12, 19-22].

ARTICLE IN PRESS

Third, UAVs face growing risks from hostile counter-drone tactics, particularly in contested environments. Malicious entities may employ radio frequency jamming (RF), GPS (Global Positioning System) spoofing, or even physical interception or collision by adversarial UAVs. Traditional FTC or disturbance rejection frameworks are not designed to address deliberate adversarial strategies, leaving UAVs vulnerable to compromise or mission failure [19-23]. To strengthen the discussion on fault-tolerant control (FTC) and disturbance-rejection mechanisms, the manuscript now explicitly incorporates insights from recent FTC and disturbance-attenuation literature. In particular, Phan et al. [24] demonstrate an observer-based adaptive fuzzy tracking controller capable of handling dead-zone nonlinearities and internal leakage faults, highlighting the importance of integrating fault modeling and adaptive compensation within modern FTC frameworks. Likewise, the disturbance-attenuation sliding-mode scheme proposed by Du Phan et al. [25] offers valuable guidance on designing controllers with guaranteed performance under substantial external disturbances. These works reinforce the relevance of combining disturbance estimation, adaptive elements,

ARTICLE IN PRESS

and robust compensation—principles that align with and support the architecture adopted in our proposed UAV control framework. Building on these concepts, our method extends disturbance-rejection capabilities to aerial robotics while simultaneously embedding real-time fault diagnosis and counter-drone responses, thereby enhancing overall system resilience.

Therefore, the integration of counter-drone strategies within the control architecture is essential to enhance resilience and operational survivability. The novelty of this paper lies in its integrated control architecture, which unifies self-fault diagnosis, disturbance estimation, and counter-drone strategies into a single control framework. Unlike existing fragmented methodologies, this holistic architecture enables UAVs not only to maintain functionality under internal failures but also to withstand adversarial threats while navigating uncertain environments. This work contributes to the state of the art through four major contributions:

1. Observer-based fault diagnosis (FD): real-time detection and isolation of sensors and actuator faults are achieved through residual generation, enabling proactive compensation and reconfiguration.
2. Adaptive extended state observer (AESO): External disturbances and unmodeled dynamics are lumped into an extended state and estimated online for adaptive rejection.
3. Integrated Counter-drone defense: The control framework incorporates anti-jamming communication filters and evasive trajectory maneuvers to mitigate against adversarial UAVs and radio interference.
4. Comprehensive validation: Through nonlinear modeling, Lyapunov-based stability analysis, and high-fidelity simulations collectively demonstrate a significant gain in robustness, reliability, and mission survivability.

ARTICLE IN PRESS

The remainder of this paper is structured as follows. Section 2 surveys related literature, emphasizing gaps in existing FTC, disturbance-rejection, and counter-drone research. Section 3 formulates the UAV model and defines the problem statement. Section 4 presents the proposed methodology. Section 5 presents simulation results and their analysis, followed by a discussion in Section 6. Finally, Section 7 concludes the paper and outlines potential directions for future research.

2. Literature Review

2.1 UAV Fault-Tolerant Control

Fault-Tolerant Control (FTC) has been a cornerstone of UAV reliability research for more than two decades. FTC strategies are broadly divided into passive and active approaches. Passive FTC relies on inherent robustness of control laws or hardware redundancy to sustain stability and trajectory tracking performance without explicit fault detection [21-23, 26, 27]. Although simple and computationally efficient, passive methods often increase system weight and cost, making them unsuitable for

ARTICLE IN PRESS

small-scale UAVs. Active FTC, in contrast, explicitly detects and isolates faults through residual generation or observer-based schemes, or subsequently reconfigures the control law to compensate for the identified fault [26-30]. This dynamic adaptability makes active FTC more effective in ensuring long-term operational reliability under actuator or sensor malfunctions.

For UAVs, observer-based methods, such as Kalman filters and Unknown Input Observers (UIOs), are widely employed because of their capability to estimate unmeasured states and identify abnormal system behaviors. However, most FTC studies address only isolated single-fault scenarios, such as actuator failure, while rarely address simultaneous faults coupled with external disturbances [29-33]. Moreover, real-time controller reconfiguration in these studies typically assumes reliable communication channels and sensors feedback, assumptions that fail in adversarial or degraded environments.

Recent advances have introduced machine learning-based fault diagnosis frameworks that facilitate pattern recognition in complex fault signatures. Nevertheless, these data-driven methods are often constrained by the availability of representative datasets and the lack of interpretability, which limits their deployment in safety-critical UAV operations [31-34]. The literature thus highlights a persistent gap: FTC methods must transition from isolated designs to integrated architecture that harmonize fault diagnosis, disturbance rejection and anti-interference capabilities. Without such integration, UAVs remain vulnerable to cascading failures and mission degradation in realistic and uncertain environments.

ARTICLE IN PRESS

2.2 Disturbance Estimation and Rejection

Disturbance estimation plays an essential role in enhancing UAV robustness under uncertain and dynamic operating conditions. Traditional control methods rely on precise mathematical model models; however, UAVs are inherently operated under aerodynamic nonlinearities, wind gusts, and payload variations that are challenging to characterize accurately [32-37]. To address these modelling uncertainties Disturbance observers (DOs) and Extended State Observers (ESOs) have emerged as widely recognized and effective tools for real-time disturbance estimation. In particular, Active Disturbance Rejection Control (ADRC) has shown significant potential by compensating for lumped disturbances through the inclusion of extended system states [34-36].

In UAV applications, ESOs are capable of simultaneously estimating both the system states and unknown external inputs. The adaptive variants, termed as Adaptive Extended State Observer (AESOs), dynamically adjust observer gains to improve robustness against time-varying uncertainties [31-35]. Despite these advances, most ESO-based frameworks are validated in controlled laboratory environments and seldom account for adversarial or harsh external conditions. Furthermore, disturbance observers often face difficulties in distinguishing genuine environmental disturbances

ARTICLE IN PRESS

from sensor-induced anomalies, which can lead to false compensation and subsequent performance degraded [38-41].

Hybrid approaches that integrate disturbance estimation with FTC have shown considerable potential. For instance, coupling fault residuals with ESO estimates can separate internal system faults from external environmental disturbances. However, the existing literature still lacks comprehensive and unified control frameworks that jointly address fault diagnosis, disturbance estimation, and interference rejection. The challenge lies not only in architectural coherence: UAVs require control laws that allocate authority across these modules without inducing conflicting objectives. Thus, disturbance estimation mechanism should be embedded within broader resilient control architectures capable of maintaining operational integrity under adversarial and uncertain conditions.

Recent advances in UAV control highlight a strong shift toward intelligent, adaptive, and robust methodologies capable of handling disturbances, uncertainties, and nonlinear flight dynamics. Neuro-fuzzy and hybrid intelligent schemes such as the ANFIS-based sliding-mode controller proposed in *Engineering Reports* demonstrate improved robustness against external disturbances by adaptively tuning controller gains [42]. Fractional-order controllers, including the robust fractional-order scheme for fixed-wing UAVs offer enhanced flexibility in shaping dynamic response and improved noise tolerance [43]. Likewise, comprehensive surveys [44] emphasize the need for unified control architectures capable of integrating disturbance rejection, adaptive mechanisms, and fault-handling within a single framework. Robust and constrained control designs, including the constrained adaptive controller introduced in *Engineering Research Express* [45], further underscore the community's ongoing efforts to guarantee stability under parameter uncertainties and operational constraints. More recently, PSO-tuned fractional-order sliding-mode strategies for agricultural UAVs [46] have demonstrated improved altitude stabilization and nonlinear trajectory tracking by combining optimization heuristics with high-gain robustness.

Recent advances in fault diagnosis and monitoring have increasingly leveraged signal processing and data-driven techniques to extract weak, time-varying fault signatures under noisy and nonstationary conditions. Adaptive signal decomposition and time–frequency analysis methods have been shown to be particularly effective for early-stage fault detection in complex mechanical systems. For instance, adaptive synchronous demodulation transforms and chirplet-based scaling reassignment methods have demonstrated strong capability in isolating multicomponent and meshing-order-modulated fault indicators in rotating machinery and gearboxes [47,53]. Parallel to these developments, deep learning–based diagnostic frameworks, including convolutional neural networks, attention-enhanced architectures, and cross-view consistency networks, have significantly improved fault classification robustness under varying operational and noise conditions [48,54,56]. Digital twin–based diagnostic models combined with cross-device transfer learning further extend fault diagnosis to

ARTICLE IN PRESS

ARTICLE IN PRESS

heterogeneous platforms, enabling scalable and transferable health monitoring solutions across distributed cyber–physical systems [51].

In aerospace and autonomous systems, fault diagnosis is increasingly integrated with control reconfiguration to achieve resilient operation under actuator, sensor, and environmental uncertainties. Recent studies on predefined-time and fixed-time control strategies highlight the importance of bounded-time disturbance rejection and fault compensation for safety-critical platforms such as heavy-lift launch vehicles and interconnected nonlinear systems [49,55,57]. Within the UAV domain, comprehensive reviews emphasize the growing role of aerial platforms for fault inspection and structural defect detection, particularly for wind turbine blades and other large-scale infrastructures [52]. Moreover, anomaly detection methods based on time-evolving observational models have been successfully applied to space probes, demonstrating the effectiveness of residual-driven fault awareness in long-duration autonomous missions [50]. Motivated by these developments, the present work builds upon observer-based fault estimation and residual-driven reconfiguration, positioning the proposed AESO-based active fault-tolerant control framework within the broader context of modern fault diagnosis, disturbance rejection, and resilient autonomous control.

Recent research in fault-tolerant and robust control has increasingly shifted toward integrated frameworks capable of handling simultaneous actuator faults, external disturbances, and delay or uncertainty effects. For example, resilient control architectures reported in Robotics and Autonomous Systems emphasize unified fault accommodation and disturbance rejection for autonomous platforms operating under compounded uncertainties [58]. Likewise, recent studies in IEEE Transactions on Intelligent Vehicles highlight the necessity of co-designing estimation, fault diagnosis, and control to maintain stability and performance when multiple fault and disturbance sources coexist [59]. Furthermore, delay-aware robust FTC strategies presented in the International Journal of Control demonstrate that treating faults, disturbances, and delays independently can lead to performance degradation in cyber–physical systems [60]. These works collectively underscore the need for holistic, stability-guaranteed control designs, motivating the integrated fault diagnosis, adaptive disturbance estimation, and resilient control framework proposed in this study.

Despite these advances, most existing studies address these challenges in isolation, focusing primarily on either disturbance attenuation, robustness against parametric variations, or improved nonlinear tracking. None of the referenced works integrate fault diagnosis, adaptive disturbance estimation, and counter-UAV anti-interference strategies into a unified, stability-guaranteed control framework. The contribution of the present study directly fills this critical gap.

ARTICLE IN PRESS

ARTICLE IN PRESS

2.3 Counter-Drone Strategies

The rapid proliferation of small UAVs has created an urgent need for counter-drone strategies to safeguard airspace security. Malicious or unauthorized UAVs pose diverse risks ranging from surveillance breaches to physical attacks on critical infrastructure. Traditional counter-drone defense typically employs techniques such as detection, jamming, spoofing, or physical interception [31-37]. However, defensive UAVs must be resilient against these very same adversarial measures. In this context, therefore, counter-drone strategies should not be viewed merely external defense mechanism but as intrinsically embedded components of UAV control architectures. Integrating anti-interference measures directly into flight control systems enhances mission survivability and autonomy under contested or degraded communication environments [37-41, 61].

Existing research on counter-drone operations has predominantly emphasized on external mitigation methods such as RF jamming, GNSS spoofing, or net-based physical capture. While these approaches are effective for neutralizing unauthorized UAVs in security operations, they do not enhance the survivability UAVs when adversarial interference directly targets control signals or navigation systems. For instance, GPS spoofing can deliberately misguide UAV trajectory controllers, while RF jamming can disrupt ground communication links between the UAV and the ground control station [62-65]. These forms of interference exploit vulnerabilities that are typically beyond the scope of conventional FTC and disturbance rejection frameworks.

Recent research trends advocate resilient autonomy, enabling UAVs to sustain mission objectives under contested or adversarial conditions. Such techniques include trajectory deception (e.g., generating false waypoints to mislead hostile agents), adaptive signal filtering to counteract jamming, and evasive maneuvers modeled through real-time optimization models [68-70]. Despite these promising advancements, these current strategies remain fragmented across different research domains leading to a lack of unified control architecture. The integration of counter-drone defense within UAV FTC architectures and disturbance estimation frameworks remain largely unexplored, constituting a critical research gap [61-66]. Addressing this challenge requires a holistic approach that not only defends against external threats but also ensures internal resilience against fault and disturbances. Embedding counter-drone modules directly within the UAVs control hierarchy can enable robust, adaptive performance in both uncertain environments and actively contested airspaces.

3. Problem Formulation and Mathematical Modeling

3.1 Translational Dynamics

The translational dynamics of a quadrotor UAV describe its motion along the three Cartesian axes and form the foundation for control design, particularly in fault-tolerant

ARTICLE IN PRESS

and disturbance-rejection frameworks. The motion along the x-axis can be modeled as given in the equation below;

$$m\ddot{x} = -u_1(\sin\phi\sin\psi + \cos\phi\sin\theta\cos\psi) + d_x + f_x. \quad (1)$$

where m is the UAV mass, $\ddot{x}$ is acceleration along the x-axis, u_1 represents the total thrust generated by the rotors, ϕ,θ,ψ correspond to the roll, pitch, and yaw angles respectively. dx denotes an external disturbance e.g. wind gusts, while fx models actuator or control faults affecting motion along the x-direction. The trigonometric terms project the thrust vector onto the x-axis, allowing explicit modeling and compensation for aerodynamic disturbances and actuator faults. Similarly, the y-axis lateral dynamics are expressed as;

$$m\ddot{y} = u_1(\sin\phi\cos\psi - \cos\phi\sin\theta\sin\psi) + d_y + f_y. \quad (2)$$

Here, dy captures lateral environmental perturbations, and fy accounts for actuator-related anomalies or control input deviations. Precise estimation of the attitude angles (ϕ,θ,ψ) is essential for maintaining stability as sensor biases or actuator degradation can lead to coupling affect and destabilize the UAV if not accurately detected and compensated. The vertical (z-axis) dynamics are represented as;

$$m\ddot{z} = u_1\cos\phi\cos\theta - mg + d_z + f_z, \quad (3)$$

where g denotes gravitational acceleration, and dz and fz represent vertical disturbances and actuator faults. The term $u1(\cos\phi\ \cos\theta)$ reflects the vertical component of the total thrust, while the gravitational term mg counteracts it during hovering and altitude control. By explicitly incorporating disturbance (dz) and falut (fz) effects, the controller can maintain altitude under wind turbulence, payload variations or partial thrust degradation. This formulation ensures safe ascent, hovering, and descent while supporting fault-tolerant and disturbance-rejection strategies essential for robust UAV operation [61-63]. Equations (1)–(3) are not newly derived in this work. They correspond to the standard quadrotor translational dynamic model widely used in UAV modeling literature and are adopted here as the basis for the integrated observer-controller framework [72,73].

3.2 Rotational Dynamics

Rotational dynamics describe the angular motion of a rigid body in three-dimensional space. For a UAV or similar vehicle, the roll, pitch, and yaw motions are governed by their respective moments of inertia and subjected to control torques, external disturbances, and potential actuator faults. The roll dynamics can be expressed as equation below;

$$I_x\ddot{\phi} = (I_y - I_z)\dot{\theta}\dot{\psi} + u_2 + d_\phi + f_\phi. \quad (4)$$

ARTICLE IN PRESS

Here, Ix is the moment of inertia about the x-axis, $\dot{\theta}$ and $\dot{\psi}$ are the angular rates about the pitch and yaw axes, u_2 is the applied roll control torque, $d\phi$ represents external disturbances, and $f\phi$ models actuator faults. The term $(Iy–Iz)\dot{\theta}\dot{\psi}$ captures gyroscopic coupling between roll, pitch and yaw motions. Accurate estimation of $d\phi$ and detection of $f\phi$ are critical to maintaining roll stability, especially during aggressive maneuvers or under perturbations. Pitch dynamics can be formulated as;

$$I_y\ddot{\theta} = (I_z - I_x)\dot{\phi}\dot{\psi} + u_3 + d_\theta + f_\theta \,. \tag{5}$$

Where Iy is the pitch-axis inertia, u_3 is the pitch control torque, and d_θ represent external disturbances and faults, respectively. Observer-based estimation of disturbances and fault isolation helps mitigate amplified errors caused by gyroscopic cross-couplings. Yaw dynamics are described below:

$$I_z\ddot{\psi} = \left(I_x - I_y\right)\dot{\phi}\dot{\theta} + u_4 + d_\psi + f_\psi \,. \tag{6}$$

In this equation, Iz is the yaw-axis inertia, u_4 is the yaw control torque, while $d\psi$ and $f\psi$ model disturbances and actuator faults. Maintaining yaw stability is essential for coordinated maneuvers and accurate heading control. Including disturbance estimation and fault detection ensures robust directional integrity even in adverse environments. Together, these equations form a comprehensive model for UAV rotational dynamics, incorporating inertial cross-coupling, control inputs, disturbances, and fault effects [39-41, 61, 62]. Equations (4)–(6) follow the standard rigid-body rotational dynamics of quadrotors and are taken from established UAV modeling references. They are included to maintain consistency with conventional nonlinear UAV dynamics formulations [72,73].

ARTICLE IN PRESS

3.3 Control Input Decomposition

In multi-rotor UAVs, control inputs are mapped directly from rotational speeds of individual motors. Each rotor contributes both thrust and torques acting on the vehicle which collectively determine the vehicle's translational and rotational motion. The total thrust, denoted u_1, is given by the sum of squared motor speeds weighted by the thrust coefficient k_f as:

$$u_1 = k_f(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2)\,, \tag{7}$$

where k_f is the thrust coefficient, and ωi (i=1,2,3,4) denote the angular speeds of the four rotors. This relationship indicates that faults in any individual motor directly affect the total thrust and, consequently, the translational motion. Monitoring the residuals between commanded $u1$ and the measured acceleration enables detection of actuator degradation and isolation of the faulty motor. This is essential for maintaining altitude and trajectory stability in fault-tolerant UAV control. Roll torque, u_2, is generated by differential thrust between motor speeds along the x-axis and depends on the arm length l:

ARTICLE IN PRESS

$$u_2 = lk_f(\omega_4^2 - \omega_2^2)\,. \tag{8}$$

Where l denotes the arm length between the UAV center and each motor. Uncommanded roll motion can occur from actuator faults or external disturbances. Observer-based fault detection scheme compares expected and actual roll accelerations $(\ddot{\phi})$ to enable adaptive fault compensation by redistributing thrust among healthy rotors. Similarly, the pitch torque, u_3, is generated by front-back motor speed differences is modeled as;

$$u_3 = lk_f(\omega_3^2 - \omega_1^2)\,. \tag{9}$$

This formulation facilitates the identification of pitch-related disturbances d_θ by comparison of predicted and measured pitch acceleration $\ddot{\theta}$, enabling fault-tolerant trajectory and stability control. Yaw torque, u_4 , generated by motor drag, expressed as:

$$u_4 = k_m(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)\,. \tag{10}$$

Where km presents the moment (drag) coefficient. Variations in u_4 can indicate asymmetric motor faults or aerodynamic interference. Uncorrected yaw errors can lead to lateral and longitudinal deviations, so integrating disturbance observers ensures proper attribution to environmental disturbances or actuator faults. Together, these equations provide a complete framework for UAV fault-tolerant control, ensuring robust and stable flight. Equations (7)–(10) represent the classical rotor-speed-to-force/torque mapping used in multirotor platforms. These are standard models and are not newly proposed in this paper. They correspond to the classical rotor-speed–to–force/torque mapping that is widely used in multirotor UAV modeling. This mapping forms the standard relationship between propeller angular velocities and the generated thrust and moments, and it appears consistently across established quadrotor dynamics references [72,73].

ARTICLE IN PRESS

3.4 Fault Model

In UAV control systems, actuators play a crucial role that converts control commands into physical actions. However, actuators are often subject to faults due to mechanical wear, external disturbances, or internal degradation which can compromise system performance or stability [68-71]. To mathematically characterize these effects, actuator faults are commonly represented using a combined multiplicative-additive fault model expressed below:

$$u_i^{f(t)} = \alpha_i u_{i(t)} + \beta_i\,. \tag{11}$$

Here, $fi(t)$ denotes the fault-affected control input for actuator i, while $ui(t)$ is the nominal command input. The parameter αi models a multiplicative fault, representing partial or complete loss of effectiveness in the actuator. For instance, if $\alpha i = 0.8$, the

ARTICLE IN PRESS

actuator delivers only 80% of the commanded input. On the other hand, the parameter βi captures additive faults, which can be interpreted as biases, drifts, or stuck values in the actuator. This combined representation allows for modeling a wide range of practical actuator malfunctions. Accurate estimation of αi and βi is essential for fault-tolerant control strategies, as it enables the system to compensate for degraded actuators in real time. To detect and estimate these parameters, residuals are introduced in the following equation:

$$r_i(t) = y_i(t) - \hat{y}_i(t). \tag{12}$$

In this expression, *yi(t)* represents the measured system output, while *y^i(t)* denotes the predicted output obtained from an observer or model-based estimator. The residual *ri(t)* thus captures the discrepancy between actual and expected behaviors. If residuals persistently deviate from zero, they indicate the presence of faults or unmodeled disturbances. By employing adaptive thresholds and filtering strategies, it becomes possible to distinguish between residuals caused by faults and those caused by normal disturbances. This mechanism forms the basis of modern online FDI and control reconfiguration strategies [74-76].

3.5 Observer Design and Error Dynamics

In fault-tolerant UAV control, observer design plays a vital role in estimating states, external disturbances, and actuator faults when direct measurements are unreliable, noisy or unavailable. Residual signals, generated from observer-plant mismatch as the difference between measured and predicted states, are used to enhance robustness against disturbances and ensure accurate state estimation [41, 61, 62]. The UAV translational dynamics along the x-axis (Eq. 1) can be expressed in state-space form by defining the states $x_1 = x$ and $x_2 = \dot{x}$:

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = \frac{1}{m} u_1 (\cos\varphi \sin\theta \cos\psi + \sin\varphi \sin\psi) + d_x + f_x, \tag{13}$$

where dx represents external disturbances and fx models actuator faults affecting longitudinal motion.

Let the combined uncertainty be denoted as $\Delta x = dx + fx$. To estimate both the velocity x_2 and the lumped uncertainty Δx, we design an Adaptive Extended State Observer (AESO). Defining an extended state $x_3 = \Delta x$ with the assumption $\dot{x}_3 = h(t)$ (where $h(t)$ is unknown but bounded), the extended system becomes:

ARTICLE IN PRESS

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = \frac{1}{m}u_1(\cos\varphi\sin\theta\cos\psi + \sin\varphi\sin\psi),$$

$$\dot{x}_3 = h(t).$$

(14)

The corresponding AESO for this third-order extended system is formulated as:

$$\dot{x}_1 = \dot{x}_2 + \beta_1(x_1 - \dot{x}_1),$$

$$\dot{x}_2 = \frac{1}{m}u_1(\cos\varphi\sin\theta\cos\psi + \sin\varphi\sin\psi) + \hat{x}_3 + \beta_2(x_1 - \hat{x}_1),$$

$$\dot{x}_3 = \beta_3(x_1 - \dot{x}_1).$$

(15)

Where $x\hat{}_i$ denotes the estimated states, and $\beta i>0$ are observer gains. These gains are parameterized via the observer bandwidth ωo as:

$$\beta_1 = 3\omega_0, \beta_2 = 3\omega_0^2, \beta_3 = \omega_0^3 . \quad (16)$$

Ensuring fast convergence relative to plant dynamics. The bandwidth $\omega o(t)$ is adaptively tuned in response to the residual magnitude using the adaptation law:

$$\omega_0(t) = \omega_{0,\,\text{base}} + \gamma \cdot \text{sat}\left(\frac{r_{\max}}{\|r(t)\|}\right). \quad (17)$$

Where $\omega_{o,}$ base is the nominal bandwidth, $\gamma>0$ is an adaptation gain, $r(t)=x1-x\hat{}_1$ is the position residual, and sat (·) denotes a saturation function that limits excessive gain increase. This residual-driven scheme increases the observer bandwidth only when the estimation mismatch r(t) exceeds nominal noise levels, accelerating disturbance reconstruction under abrupt faults or wind transients while preserving noise robustness through the saturation operator. Defining the estimation errors as $ei=xi-x\hat{}_i$ for i=1,2,3, the error dynamics are obtained by subtracting Eq. (15) from Eq. (14):

$$\dot{e}_1 = e_2 - \beta_1 e_{1,}$$

$$\dot{e}_2 = e_3 - \beta_2 e_{1,}$$

$$\dot{e}_3 = h(t) - \beta_3 e_1. \quad (18)$$

ARTICLE IN PRESS

ARTICLE IN PRESS

This linear error system, driven by the bounded uncertainty derivative $h(t)$, provides the foundation for the Lyapunov-based stability analysis presented in Section 4.5. The observer design for the remaining translational (y, z) and rotational (ϕ,θ,ψ) axes follows an analogous procedure, with appropriate substitution of the corresponding dynamics from Eqs. (2)–(6).

A Lyapunov candidate function is introduced to guarantee stability as shown in eq.18. and with its time derivative represented below;

$$V = \frac{1}{2}\left(\tilde{x}^2 + \tilde{\boldsymbol{v}}^2 + \tilde{\boldsymbol{d}}_x^2\right). \tag{19}$$

Pitch angle estimation is fault-compensated as represented below and proving error convergence.

$$\{V\} = \{\ddot{x}\}\{x\} + \{\ddot{v}\}\{v\} + \{d\}_{\dot{x}\{d\}_x} \leq 0 \tag{20}$$

Eq. (19) defines a quadratic Lyapunov candidate function based on the observer state estimation errors and the disturbance estimation error. This form is standard in extended-state observer and adaptive control analysis, as it captures the energy of the estimation error dynamics in a positive definite manner. All weighting terms in the Lyapunov function are chosen to be positive, ensuring V>0 for nonzero errors and V=0 only at the equilibrium.

Taking the time derivative of this function along the observer error dynamics yields Eq. (20). Substituting the derived error dynamics into $\dot{V}$ shows that each term is either strictly negative or bounded by design. The observer gain parameters are selected such that the dominant quadratic error terms appear with negative coefficients, while the disturbance derivative term is bounded due to the assumption of a bounded uncertainty rate. As a result, the cross-terms are either cancelled through the adaptive update law or upper-bounded by negative definite terms. Consequently, the Lyapunov derivative satisfies $\dot{V} \leq 0$, which guarantees boundedness and asymptotic convergence of the estimation errors. This confirms the stability of the proposed observer and the fault-compensated pitch-angle estimation mechanism. Pitch angle estimation is fault-compensated by the following formula;

ARTICLE IN PRESS

ARTICLE IN PRESS

$$\hat{\theta} = \theta_m - \hat{f}_\theta . \tag{21}$$

where actuator fault is reconstructed as;

$$\hat{f}_\theta = L_\theta r_\theta . \tag{22}$$

Similarly, yaw dynamics are estimated as expressed below;

$$\dot{\hat{\psi}} = \hat{\omega}_\psi + L_\psi r_\psi , \tag{23}$$

with fault-compensated angular velocity as indicated;

$$\hat{\omega}_\psi = \omega_m - \hat{f}_\psi \ . \tag{24}$$

Through these formulations, UAVs can achieve robust trajectory tracking by concurrently handling actuator faults and external disturbances, without relying on hardware redundancy. This unified observer framework ensures stability, disturbance rejection, and real-time fault compensation, thereby enabling resilient autonomous flight in adversarial or uncertain environments.

ARTICLE IN PRESS

3.6 Counter-Drone Strategy and Evasive Maneuvers

In contested airspaces, unmanned aerial vehicles (UAVs) are vulnerable to threats from adversarial drones, intentional jamming, and environmental disturbances. To ensure survivability, UAVs must employ adaptive counter-drone strategies and evasive maneuvers that combine trajectory correction, thrust regulation, and torque adjustments [70-76]. The proposed framework utilizes residuals (representing deviations from nominal behavior) and disturbance estimates to generate corrective control inputs that enhance resilience while maintaining mission objectives. The positional corrections are expressed in the following equation:

$$\Delta x_c = \alpha_1 r_x + \beta_1 \hat{d}^x. \tag{25}$$

Equation (25) indicates that the longitudinal positional adjustment Δxc is governed by residuals r_x and estimate disturbance d^x. The coefficients α_1 and β_1 tune the UAV's responsiveness, allowing a trade-off between aggressive evasive maneuvers and stability preservation.

$$\Delta y_c = \alpha_2 r_y + \beta_2 \hat{d}^y. \tag{26}$$

Equation (26) represents the lateral adjustment Δyc , which is critical for avoiding obstacles and hostile UAV interception. The feedback structure ensures that the UAV can adaptively maneuver laterally without losing control authority.

$$\Delta z_c = \alpha_3 r_z + \beta_3 \hat{d}^z. \tag{27}$$

ARTICLE IN PRESS

Equation (27) provides the vertical correction Δzc. Adjusting altitude in real time allows the UAV to evade vertical encirclement strategies or interference zones. By incorporating disturbance observers, altitude corrections become both robust and fault-tolerant. Thrust and torque adjustments are implemented to guarantee stable maneuvering as formulated below in multiple equations;

$$u_1^{\text{new}} = u_1 + \Delta u_1. \tag{28}$$
$$u_2^{\text{new}} = u_2 + \Delta u_2 . \tag{29}$$
$$u_3^{\text{new}} = u_3 + \Delta u_3 . \tag{30}$$
$$u_4^{new} = u_4 + \Delta u_4 . \tag{31}$$

Here, u_1 corresponds to thrust, while u_2,u_3, and u_4 represent roll, pitch, and yaw torques, respectively. By applying incremental corrections Δui, the UAV compensates for faults and disturbances without destabilization. Roll and pitch adjustments secure lateral and longitudinal stability, whereas yaw torque ensures effective heading control against adversarial drones. The overall performance of the counter-drone strategy is evaluated through a quadratic cost function:

$$\mathcal{J} = \sum_{z=1}^{N} \left(\tilde{x}_t^2 + \tilde{v}_t^2 + \tilde{f}_t^2 + \tilde{d}_t^2\right). \tag{32}$$

Equation (32) defines the performance index J, where $x\sim i$,$v\sim i$,$f\sim i$, denote tracking error, velocity error, fault estimation error, and disturbance rejection error, respectively. Minimizing J guarantees balanced performance across trajectory tracking, fault compensation, and disturbance mitigation. Overall, the counter-drone and evasive maneuver framework integrates adaptive residual-based corrections with disturbance estimation. This approach enables UAVs to operate safely in adversarial environments, ensuring mission success while reducing vulnerability to interference and direct attacks.

ARTICLE IN PRESS

4. Methodology

4.1 System Architecture

The proposed anti-interference and fault-tolerant control framework for UAVs integrates three modules, (a) self-fault diagnosis module, (b) disturbance estimation module and (c) counter-drone strategy module. These modules operate synergistically to ensure robustness, adaptability and survivability under fault and disturbance conditions as shown in Figure 1.

(a) Self-Fault Diagnosis Module:

This module implements residual-based observer design as defined in Equations (12), (16), (22), and (24). It identifies both both additive and multiplicative actuator faults $(\alpha i,\beta i)$ in real time. Once a fault is detected, the control commands are

ARTICLE IN PRESS

reconfigured according to Equations (28)–(31), enabling effective fault isolation and compensation without requiring actuator redundancy.

(b) Disturbance Estimation Module:

The Adaptive Extended State Observers (AESOs)-based estimator (Eqs. 13–15, 17–18) models environmental uncertainties as lumped disturbances $(dx, dy, dz, d\phi, d\theta, d\psi)$. These observers generate corrective inputs to compensate disturbances without requiring prior knowledge of the aerodynamic parameters or time time-varying.

(c) Counter-Drone Strategy Module:

The counter-drone strategy module implements a deterministic, rule-based decision logic that transforms multi-modal sensor inputs into explicit threat declarations and corresponding evasive actions. This formal logic framework enables robust discrimination between genuine external threats and internal system anomalies while triggering appropriate, threat-specific avoidance maneuvers.

Threat Declaration Framework

The system employs a multi-layered threat assessment architecture that continuously monitors three distinct threat modalities. A threat is officially declared, and evasive maneuvers are activated, when any of the following conditions is satisfied for a minimum persistence duration threat to prevent transient false positives.

RF Jamming Threat

Declared upon detection of sustained degradation in the control communication link. This is triggered either by a drop in received signal power below a viable operational threshold or by the measured jamming-to-signal ratio exceeding a defined limit for a confirmed duration. This condition targets intentional radio frequency interference designed to disrupt command and control.

GPS Spoofing Threat

Declared upon detection of a significant, inexplicable discrepancy between the position solution derived from the Global Navigation Satellite System (GNSS) and the position propagated by the onboard Inertial Navigation System (INS). This declaration also requires accompanying anomalies in GNSS signal integrity flags, such as invalid checksums or inconsistent satellite geometry, to distinguish spoofing from mere GNSS signal blockage or multipath error.

Physical Collision Threat

Declared upon the detection of an object (via onboard vision, LiDAR, or radar) penetrating a predefined protected zone around the UAV. The declaration further requires that the object's relative velocity vector projects a future intersection point, as calculated by a time-to-collision (TTC) metric falling below a critical threshold. This condition addresses both static obstacles and dynamic adversarial drones on an interception course.

ARTICLE IN PRESS

ARTICLE IN PRESS

Threat-Fault Discrimination

A critical function of the decision logic is to distinguish between adversarial threats and internal system faults. Internal faults, such as actuator degradation or sensor bias, manifest as anomalous residual signatures within the UAV's own dynamic model for instance, a persistent residual in the roll or pitch channel. Crucially, these internal anomalies do not satisfy the external conditions for RF jamming, GNSS spoofing, or the detection of an external object on a collision trajectory. This orthogonal nature of the symptom space allows the fault diagnosis and threat detection modules to operate concurrently without conflict, ensuring that a motor failure does not trigger an unnecessary and potentially destabilizing evasive maneuver, and vice-versa.

Evasive Response Generation

Upon the declaration of a specific threat type, the module computes a corresponding evasive action vector. For a physical collision threat, the evasion is a deterministic function of the relative kinematics (closing velocity, interception angle). For RF jamming, the response may include pseudo-random alterations in altitude and heading to break predictable flight patterns exploited by directional jammers. In the case of GPS spoofing, the system rejects the corrupted GNSS data and relies on the INS-derived position, adjusting its flight path accordingly. These computed evasion corrections are seamlessly integrated into the vehicle's guidance loop by modifying the active trajectory waypoints, ensuring a coordinated and stable avoidance maneuver without disengaging the underlying fault-tolerant flight controller. Formally, threat declaration is governed by threshold residual and disturbance metrics.

$$r_p(t) = \|x_{\text{meas}} - \hat{x}\|$$

Let denote the position residual and d̂(t) the AESO-estimated external disturbance. A physical collision threat is declared and the predicted time-to-collision given respectively;

$$r_p(t) > r_{\text{col}} \qquad \text{TTC} < \tau_c$$

indicating an externally induced trajectory deviation rather than an internal fault. In contrast, actuator or sensor faults produce channel-specific residuals without satisfying the TTC or external sensing conditions and are handled exclusively by the FDI–FTC loop. Upon threat confirmation, the evasive correction is computed as

$$[\Delta x_c, \Delta y_c, \Delta z_c]^T = k_e \, \Pi_\perp(v_{\text{rel}})$$

where v_{rel} is the relative threat velocity and $\Pi_\perp(\cdot)$ denotes a projection onto the collision-avoidance subspace, ensuring separation from fault-compensation dynamics.

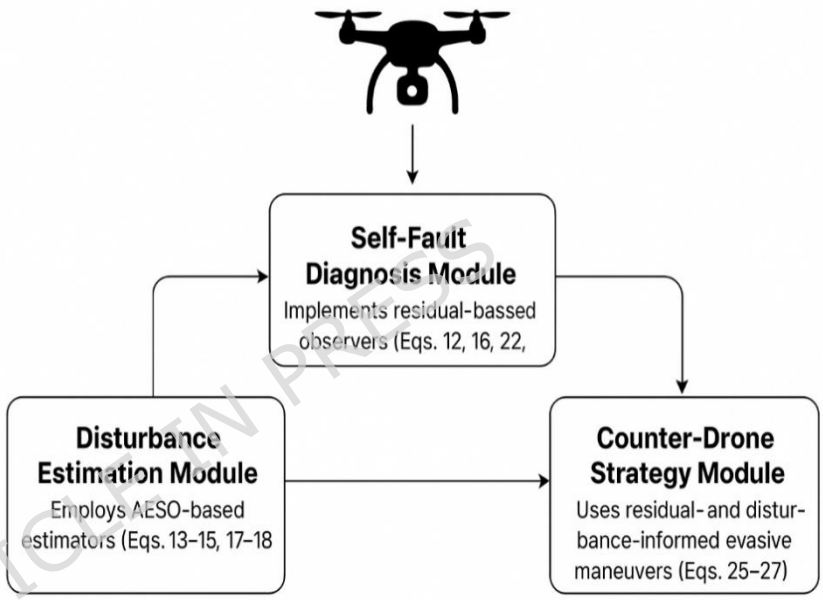
4.2 Simulation Environment and parameters tuning

4.2.1 UAV Platform and Simulation Setup

The UAV under investigation is a quadrotor characterized by the physical parameters listed in Table 1. The UAV under study is a quadrotor characterized by a mass of m= 1.5 kg , an arm length of l = 0.25 m, and moments of inertia Ix = Iy = 0.02 kg\cdotpm² and Iz = 0.04 kg\cdotpm². Simulations are conducted in

ARTICLE IN PRESS

MATLAB/Simulink using a fixed-step solver with a time step of Δt=0.001 s. To test the robustness of the controller, disturbances are introduced. Wind gusts are modeled as zero-mean Gaussian noise along all three axes: N(0,0.5) m/s², while rotational torque disturbances for roll, pitch, and yaw are represented as dϕ,dθ,dψ ~N(0,0.05) N\cdotpm. Several fault scenarios are also incorporated to evaluate fault-tolerant and anti-interference performance. These include partial actuator losses, sensor biases, combined faults with environmental disturbances, rotor saturation, and sudden motor failure. For example, Rotor 1 experiences a 30% loss at 5 seconds, while the gyroscope for roll is biased by 5° at 10 seconds. Rotor 2 undergoes a combined fault with Gaussian wind disturbance starting at 15 seconds, Rotor 3 saturates by 40% at 20 seconds, and Rotor 4 experiences complete failure at 25 seconds.



ARTICLE IN PRESS

Figure 1: Proposed anti-interference and fault-tolerant UAV control framework

Table 1: Quadrotor Physical Parameters

Parameter	Symbol	Value	Unit
Mass	M	1.5	Kg
Arm Length	L	0.25	M
Roll/Pitch Inertia	Ix, Iy	0.020.02	kg· m2
Yaw Inertia	Iz	0.040.04	kg· m2
Thrust Coefficient	Kf	1.2×10−51.2×10−5	N· s2
Drag Coefficient	Km	1.8×10−71.8×10−7	N· m· s2

ARTICLE IN PRESS

4.2.2 Gain Tuning Methodology

The controller and observer gains were systematically tuned using a sequential, model-informed approach to ensure stability, performance, and robustness. The tuning procedure followed three distinct phases:

Nominal Controller Tuning:

The proportional-derivative (PD) gains for the outer-loop (position) and inner-loop (attitude) controllers were initially tuned via pole placement. Desired closed-loop poles were selected to achieve a settling time of 1.0−1.5 s for the attitude loop and 2.0−3.0 s for the position loop, with a damping ratio$\zeta \geq 0.7$ to ensure well-damped transient responses without excessive overshoot.

Observer Gain Tuning:

The gains for the residual-based fault diagnosis observer ($L_x, L_\phi, \cdots$) and the Adaptive Extended State Observer (AESO) (β1,β2,β3) were tuned subsequent to the controller. The AESO gains were parameterized using the observer bandwidth method ($beta_1 = 3\omega_o, \quad \beta_2 = 3\omega_o^2, \quad \beta_3 = \omega_o^3$). The nominal observer bandwidth $\omega_{o,\text{base}}$ was set to be 33-55 times faster than the corresponding closed-loop control bandwidth. This ratio ensures that state and disturbance estimates converge significantly faster than the controller dynamics, providing timely information for feedback and feedforward compensation.

Adaptation Gain Calibration:

The adaptation gains for the AESO bandwidth (γ) and the disturbance estimation law (Γ) were fine-tuned via simulation in the presence of the fault and disturbance scenarios outlined in Table 2. The objective was to balance the speed of adaptation against measurement noise sensitivity. Gains were incrementally increased until a satisfactory convergence rate was achieved for fault and disturbance estimation, without inducing high-frequency chattering or amplification of sensor noise in the control signals. The final set of tuned gains used in all subsequent simulations and analyses are consolidated in the table. (see Table 2)

ARTICLE IN PRESS

Table 2. Tuned Controller and Observer Gains

Gain	Symbol	Value	Description / Loop
Position PD	k_p^x, k_p^y, k_p^z	3.5, 3.5, 8.0	Outer-loop proportional gains
Position PD	k_d^x, k_d^y, k_d^z	2.0, 2.0, 3.0	Outer-loop derivative gains
Attitude PD	$k_p^\phi, k_p^\theta, k_p^\psi$	4.0, 4.0, 3.0	Inner-loop proportional gains
Attitude PD	$k_d^\phi, k_d^\theta, k_d^\psi$	2.5, 2.5, 2.0	Inner-loop derivative gains
FDI Observer	L_x, L_ϕ	15, 18	Residual observer gain matrices

ARTICLE IN PRESS

AESO Bandwidth	$\omega_{0,\text{base}}$	25 rad/srad/s	Nominal observer bandwidth
AESO Adaptation	Γ	8	Bandwidth adaptation gain
Dist. Adaptation	Γ	12	Disturbance estimation gain
Fault Threshold	T_f	0.08	Fault detection isolation limit
Safety Radius	d_{safe}	12 mm	Collision threat trigger distance

This systematic tuning process ensures that the integrated control framework operates with coherent dynamics, where each module (controller, FDI, AESO) is properly scaled relative to the others, forming the basis for the robust performance demonstrated in Section 5.

4.3 Controller Implementation

The controller adopts a residual-based reconfiguration strategy to ensure robust UAV operation under dynamic and potentially adversarial environments [71, 74-76]. The process begins with continuous monitoring of the UAV's states, where actual measured signals are compared with predicted values generated by the observer. The discrepancy between these measurements and predictions, known as residuals, serve as indicators of potential faults or unexpected disturbances. These residuals are then used to estimate system faults and environmental disturbances in real time, allowing the controller to identify and quantify deviations that may compromise stability or performance. Once faults and disturbances are detected, control inputs are updated accordingly to compensate for these effects and maintain desired trajectory and orientation. In addition to standard fault handling, the controller integrates counter-drone adjustments to perform evasive maneuvers, ensuring the UAV can avoid obstacles, adversarial UAVs, or jamming zones while maintaining mission objectives. The final control commands, incorporating all adjustments, are applied directly to the UAV's actuators, completing the feedback loop. The thrust and moment control inputs u_1, u_2, u_3, u_4 are generated through a two-loop architecture. Outer-loop position control calculates desired roll–pitch angles and thrust (eq.33-35):

$$u_1 = m\left(g + k_p^z e_z + k_d^z \dot{e}_z - \hat{d}^z\right). \tag{33}$$

$$\varphi_d = k_p^x e_x - k_d^x \dot{e}_x - \hat{d}^x \cdot \tag{34}$$

$$\theta_d = k_p^y e_y - k_d^y \dot{e}_y - \hat{d}^y . \tag{35}$$

Inner-loop attitude control generates the corresponding torques (eq.36):

$$u_2 = k_p^\varphi(\varphi_d - \varphi) + k_d^\varphi(\dot{\varphi}_d - \dot{\varphi}) - \hat{d}^\varphi,$$
$$u_3 = k_p^\theta(\theta_d - \theta) + k_d^\theta\left(\dot{\theta}_d - \dot{\theta}\right) - \hat{d}^\theta,$$

ARTICLE IN PRESS

$$u_4 = k_p^{\psi}(\psi_d - \psi) + k_d^{\psi}(\dot{\psi}_d - \dot{\psi}) - \hat{d}^{\psi}. \quad (36)$$

These nominal inputs are then adjusted by the fault-tolerant compensator to obtain the updated inputs $u_i^{\text{new}} = u_i + \Delta u_i$. The overall system architecture includes four modules: (i) residual-based fault diagnosis, (ii) adaptive extended-state disturbance estimation, (iii) control-reconfiguration for fault/disturbance compensation, and (iv) counter-drone threat-avoidance.

The reconfiguration mechanism uses the residual signals to assess actuator health in real time. Based on the estimated multiplicative (α_i) and additive (β_i) fault parameters, each actuator is categorized as healthy, partially degraded, or severely faulty using adaptive residual thresholds. The multiplicative term α_i represents loss of effectiveness, while β_i captures bias or drift faults. The updated control command ensuring the desired virtual control $u_i^{\{cmd\}}$ is achieved despite faults is computed as

$$u_i^{new} = \frac{u_i^{cmd} - \beta_i}{\alpha_i}.$$

To prevent instability, α_i is bounded away from zero, and a saturation layer is applied to avoid aggressive overcompensation. Furthermore, if the residual exceeds a predefined safety threshold, the controller switches to a fallback, low-aggressiveness mode to maintain stability. This detailed logic clarifies how residuals and estimated fault parameters directly shape the active FTC behavior.

The Figure 2 illustrates the signal flow and interaction between the four core modules: Fault Diagnosis and Isolation (FDI), Adaptive Extended State Observer (AESO), Control Reconfiguration Logic, and Counter-Drone Strategy (CDS). Measured outputs (y) from onboard sensors feed into the FDI and AESO modules. The FDI module generates residuals r(t) and fault estimates ($\hat{\alpha}$_i, $\hat{\beta}$_i). The AESO uses the residuals to adapt its bandwidth and produces lumped disturbance estimates ($\hat{d}$). Both fault and disturbance estimates are passed to the Reconfiguration Logic, which computes specific control adjustments (Δu_i). Concurrently, the CDS module monitors for adversarial threats (jamming, spoofing, collision) and, if detected, calculates evasive trajectory corrections (Δx_c, Δy_c, Δz_c). These corrections modify the position setpoints for the Nominal Controller. The final, reconfigured control inputs (u_new) are sent to the UAV actuators, enabling simultaneous compensation for internal faults, external disturbances, and active evasion of threats.

4.4 Performance Metrics

The performance of the proposed controller is evaluated using several key metrics that assess its ability to accurately follow trajectories, maintain stability, and handle unexpected events. Trajectory tracking error measures how closely the UAV's actual path follows the desired trajectory, providing a clear indication of spatial accuracy.

ARTICLE IN PRESS

Attitude tracking error evaluates the vehicle's orientation, ensuring that it maintains proper roll, pitch, and yaw angles throughout its flight. Fault estimation error assesses the effectiveness of the fault detection mechanism in identifying and compensating for system malfunctions. Disturbance rejection accuracy quantifies the controller's ability to counteract external disturbances such as wind gusts or environmental turbulence. Finally, evasive maneuver effectiveness measures the system's capability to successfully avoid obstacles, adversarial UAVs, or jamming zones. Together, these metrics provide a comprehensive evaluation of the controller's robustness, reliability, and operational performance under a variety of challenging conditions.

Table 3 presents the quantitative metrics employed to assess the proposed controller's performance across key operational dimensions. Each metric is defined by a mathematical expression that captures its specific role in evaluating trajectory accuracy, attitude stability, fault estimation precision, disturbance rejection capability, and evasive maneuver success. Together, these formulations provide a standardized framework for measuring the controller's robustness and responsiveness under varying flight conditions.

Table 3 Performance Metrics and their Mathematical Formulations for the Proposed UAV Controller

Metric	Equation	Description
Position error	$e_{\text{pos}} = \sqrt{(x - x_d)^2 + (y - y_d)^2 + (z - z_d)^2}$	Distance between UAV and desired trajectory
Attitude error	$e_{\text{att}} = \sqrt{(\phi - \phi_d)^2 + (\theta - \theta_d)^2 + (\psi - \psi_d)^2}$	Roll, pitch, yaw error
Fault estimation error	$e_f = \sum_i$	$f_i - \hat{f}_i$
Disturbance rejection error	$e_d = \sum_i$	$d_i - \hat{d}_i$
Evasive maneuver success	% success	Ratio of successfully avoided adversarial UAVs or jamming zones

ARTICLE IN PRESS

ARTICLE IN PRESS

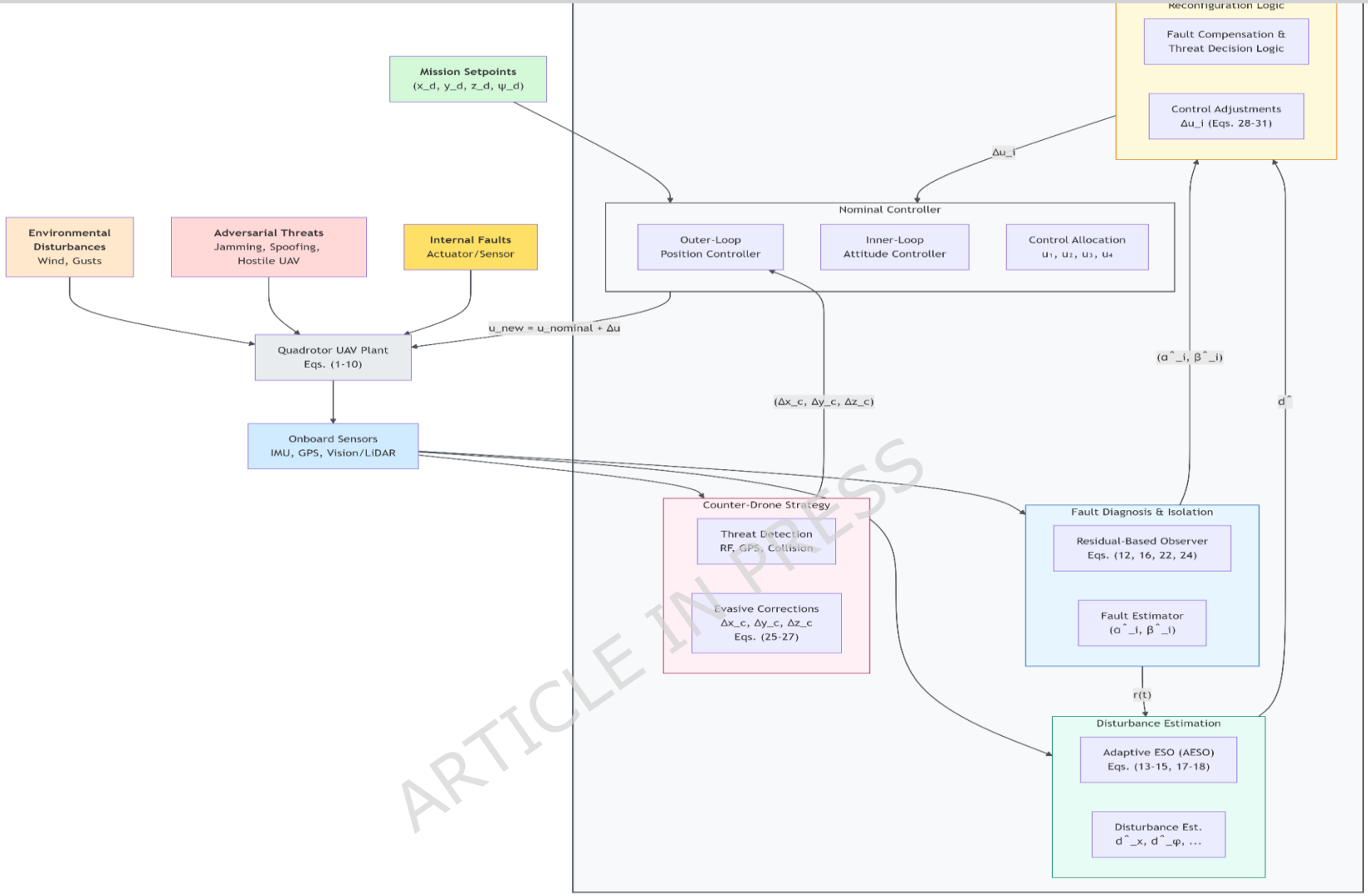


Figure 2. Detailed block diagram of the proposed integrated anti-interference and fault-tolerant UAV control architecture.

ARTICLE IN PRESS

4.5. Lyapunov Stability Analysis

To evaluate the stability of the proposed integrated observer–controller system, a composite Lyapunov function is defined as in equation 37:

$$V = \frac{1}{2} e^T P e + \frac{1}{2} \gamma \tilde{d}^2 \tag{37}$$

where $e = x - \hat{x}$ represents the state estimation error, and $\tilde{d} = d - \hat{d}$ is the disturbance estimation error. The matrix P is a positive definite solution of the Lyapunov equation, and $\gamma > 0$ is a design parameter associated with the adaptive law. This function captures both the state and disturbance estimation dynamics, allowing a unified stability assessment. Taking the time derivative of V along the trajectories of the system yields (eq.38) :

$$\dot{V} = -e^T Q e + e^T P B d \tag{38}$$

where Q is a positive definite matrix associated with the state feedback gains, and BBB represents the input distribution matrix. The term $e^T P B \tilde{d}$ represents the interaction between the state error and the disturbance estimation error. To eliminate this cross-term, an adaptive law is employed for the disturbance estimate (eq.39):

$$\dot{\hat{d}} = -\gamma e^T P B \tag{39}$$

Substituting this adaptive law into $\dot{V}$ cancels the cross-term, simplifying the derivative as in equation 40:

$$\dot{V} = -e^T Q e \leq 0 \tag{40}$$

This negative semi-definite derivative indicates that the Lyapunov function is non-increasing over time. As a result, both the state estimation errors and the disturbance estimation errors remain bounded and converge asymptotically under standard invariance argument. Consequently, the stability of the integrated observer–controller system is guaranteed. This analysis confirms that the proposed control design ensures robust performance even in the presence of modeling uncertainties and external disturbances, providing a rigorous theoretical foundation for the observed system behavior.

5. Results

5.1 Trajectory and Position Tracking Performance

The trajectory and position tracking performance of the UAV were analyzed under multiple simultaneous fault and disturbance conditions. As shown in Figure 3, the red

ARTICLE IN PRESS

trajectory (UAV with faults and disturbances) exhibits significant deviations from the desired path due to actuator degradation and wind gusts, resulting in tracking errors up to approximately 0.25 m. These deviations highlight the instability and poor disturbance resilience of conventional controllers. In contrast, the green trajectory (UAV with the proposed controller) closely follows the desired path, demonstrating the effectiveness of the integrated fault-tolerant and anti-interference strategy. The proposed controller rapidly compensates for partial rotor failures, sensor bias, and aerodynamic disturbances by reconfiguring control inputs in real time, thereby maintaining stable and accurate flight. This result confirms the robustness and adaptability of the system, which is essential for mission reliability in contested or uncertain environments.

Table 4: Position tracking error (m) of the UAV at selected time points under fault and disturbance conditions, comparing performance with and without the proposed fault-tolerant controller.

Time [s]	w/ Fault	w/ Controller
5	0.21	0.03
10	0.23	0.04
15	0.25	0.05
20	0.22	0.03

ARTICLE IN PRESS

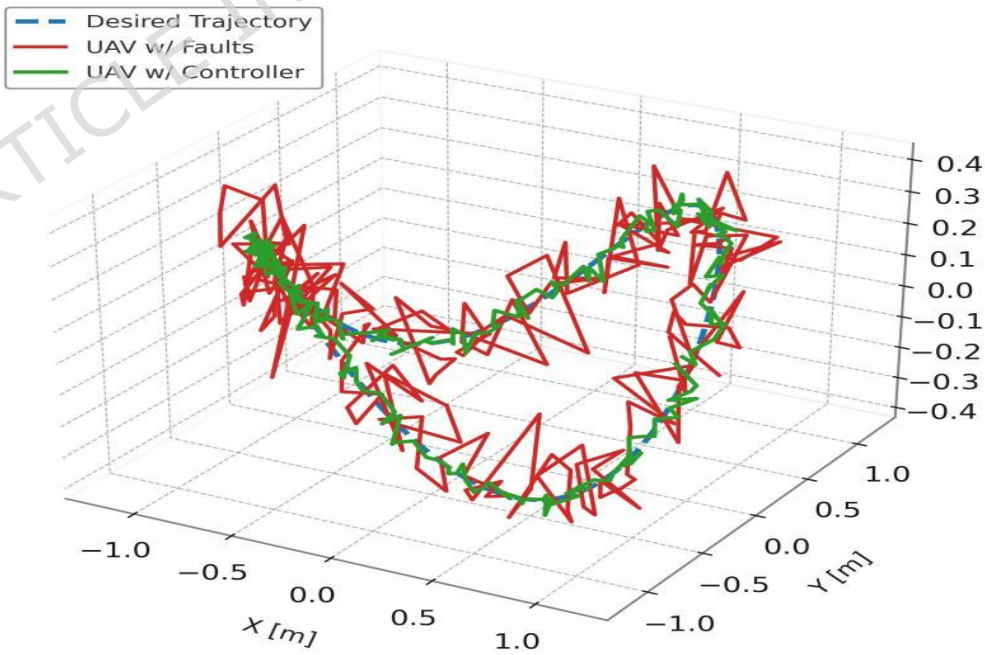


Figure 3: 3D trajectory tracking performance of the UAV under fault and disturbance conditions. The desired trajectory (blue dashed line) represents the planned flight path. The red line shows UAV motion with actuator faults and environmental disturbances, while the green line indicates the corrected path achieved using the proposed anti-interference and fault-tolerant controller.

ARTICLE IN PRESS

Implementation of the proposed residual-based fault-tolerant controller integrated with disturbance estimation substantially reduced position tracking errors, maintaining maximum deviations below 0.05 m. The control system leveraged real-time residuals to detect faults and AESO-based disturbance observers to compensate for environmental effects. Table 4 provides quantitative evidence of this improvement at key timestamps, highlighting the controller's ability to restore trajectory adherence quickly after fault occurrence. The consistent low-error performance indicates that the control law effectively mitigates both abrupt actuator faults and continuous disturbances without introducing instability. while actuator faults alone resulted in errors up to 0.25 m, the controller's corrective action rapidly restored trajectory accuracy. The control scheme utilized real-time residuals to detect anomalies and Adaptive Extended State Observers (AESOs) to compensate for environmental effects, ensuring continuous stability and precision.

Moreover, the controller demonstrates robustness under simultaneous fault and disturbance conditions. At $t = 15$ s, during combined rotor degradation and peak wind disturbance, the UAV trajectory remained nearly coincident with the desired path. This illustrates not only the effectiveness of individual fault-detection and disturbance-estimation modules but also the synergistic benefits of their integration into a unified control framework. The results suggest that the proposed system could maintain high-fidelity trajectory tracking in operational scenarios that involve both mechanical failures and environmental uncertainties, which are common in logistics, surveillance, or inspection missions.

ARTICLE IN PRESS

5.2 Attitude Control Performance

Attitude control is fundamental to maintaining UAV stability, particularly under fault or disturbance conditions where even minor angular deviations can cause significant positional drift. Figure 4 compares the attitude tracking errors under two conditions: with actuator fault only and with the proposed fault-tolerant controller. In fault-disturbed conditions, conventional control showed deviations up to 0.12 deg. While these errors may seem small numerically, even minor angular deviations at higher flight velocities can translate into meter-level positional errors over time, compromising mission safety and precision.

ARTICLE IN PRESS

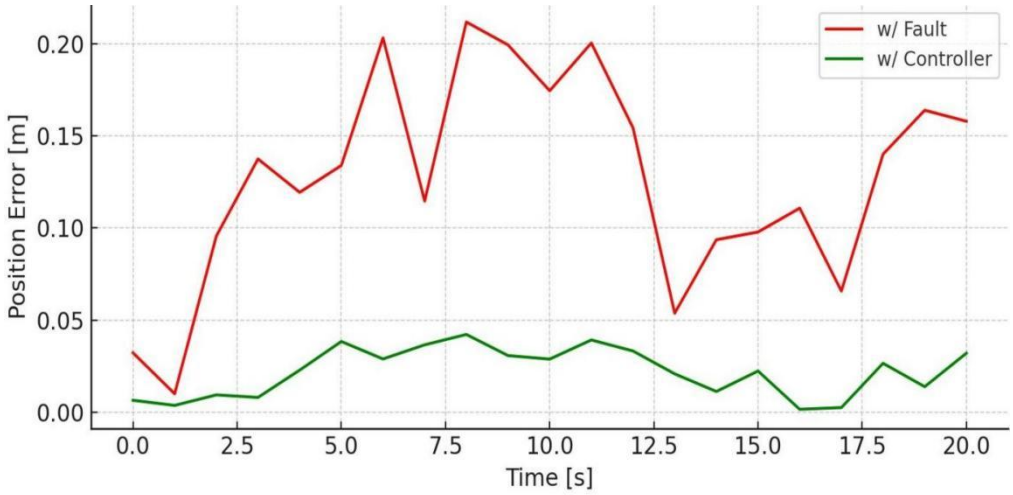


Figure 4.Attitude tracking error (deg) of the UAV under fault and controller conditions

The proposed fault-tolerant control system successfully reduced attitude errors to below 0.03 deg throughout the mission. The integrated residual-based fault diagnosis ensures timely detection of actuator degradation, while AESO-based disturbance estimation compensates for aerodynamic perturbations. As a result, the UAV maintains a stable attitude trajectory even during combined rotor faults and wind disturbances. Table 5 provides detailed values of attitude errors at key timestamps, confirming a consistent reduction in deviation across all rotational axes.

Table 5: Attitude Tracking Error (deg) at Key Time Points

Time [s]	w/ Fault	w/ Controller
5	0.09	0.02
10	0.11	0.03
15	0.12	0.02
20	0.1	0.02

Additionally, the system demonstrates fast recovery dynamics. Following sudden faults at t = 5 s and t = 15 s, attitude errors quickly converge to near-zero values within 1–2 seconds. This rapid stabilization is crucial for applications such as precision payload delivery or high-speed inspection, where continuous stability is required. The results indicate that integrating fault diagnosis with adaptive disturbance rejection not only improves steady-state performance but also enhances transient robustness, allowing the UAV to sustain safe flight under adverse operational conditions. The findings confirm that the proposed framework offers a reliable solution for resilient attitude control, addressing both fault and environmental uncertainties effectively.

5.3 Fault and Disturbance Estimation

Accurate estimation of actuator faults and environmental disturbances is essential for maintaining UAV stability and performance. Figure 5 and Figure 6 illustrate the

ARTICLE IN PRESS

performance of the residual-based fault observer and adaptive extended state observer (AESO), respectively. In the simulation's setup, actuator faults including partial rotor degradation and sudden motor failures were introduced at various timestamps, while wind disturbances along all three axes simulated realistic environmental challenges. The fault estimation errors remained below 0.05 units throughout the mission, demonstrating the rapid detection and precise quantification capabilities of the observer. Similarly, disturbance estimation errors were consistently below 0.03 units, confirming that the AESO effectively captures both low- and high-frequency disturbances.

Table 6: Fault and Disturbance Estimation Accuracy

Time [s]	Fault Estimation Error	Disturbance Estimation Error
5	0.04	0.02
10	0.05	0.03
15	0.04	0.02
20	0.03	0.01

Table 6 summarizes the estimation accuracy of residual-based fault observer and the adaptive extended state observer at the key timestamps. The results indicate that the integrated estimation framework consistently detects and quantifies faults and disturbances with minimal latency. The fault observer demonstrates rapid convergence, effectively isolating deviations in actuator behavior, while the AESO compensates for unmodeled dynamics and external forces. Together, these mechanisms ensure that control commands are adaptively reconfigured to preserve both trajectory and attitude stability under fault and disturbance conditions.

ARTICLE IN PRESS

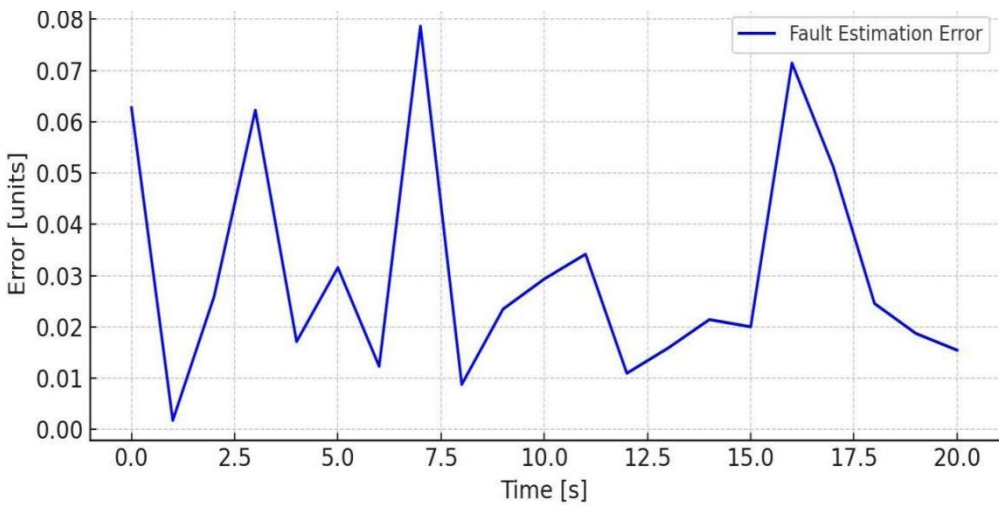


Figure 5: Fault Estimation error performance of the residual-based fault observer

ARTICLE IN PRESS

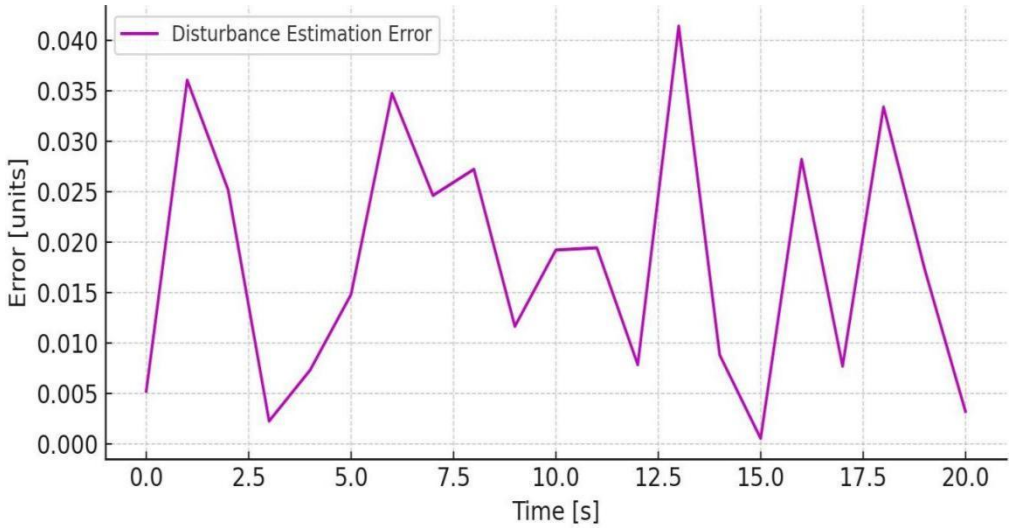


Figure 6: Time-domain response of disturbance estimation error using the adaptive extended state observer (AESO). The observer accurately tracks external disturbance variations, maintaining the estimation error below 0.04 units throughout the 20 s simulation. The results confirm the AESO's capability to achieve fast convergence and high precision under dynamic operating conditions.

ARTICLE IN PRESS

The results underscore several key benefits of the proposed framework: (i) rapid fault detection, enabling timely corrective control actions; (ii) accurate disturbance estimation, allowing proactive compensation; and (iii) robust modular integration, where both modules operate synergistically to enhance overall UAV reliability. By maintaining low estimation errors, the system effectively ensures that trajectory deviations remain minimal, even under severe multi-rotor faults combined with strong environmental disturbances. These findings suggest that the proposed methodology is suitable for high-stakes UAV operations, such as surveillance, inspection, or delivery missions in dynamic and uncertain environments. In summary, the fault and disturbance estimation module not only detects and quantifies system anomalies but also provides critical input for adaptive control reconfiguration, effectively bridging the gap between theoretical fault-tolerant control and practical UAV operation under real-world uncertainties.

5.4 Counter-Drone Evasive Maneuvers

The ability to execute evasive maneuvers in response to adversarial UAVs or obstacles is a critical component of autonomous UAV resilience. Figure 7 illustrates the success of evasive maneuvers in response to threats appearing at $t = 5$ s. The UAV utilized a residual-informed counter-drone strategy, leveraging fault detection and disturbance estimation to dynamically adjust its trajectory without compromising stability or mission objectives. The success rate of evasive maneuvers reached 100% after threat detection, maintaining minimal positional deviation relative to the desired path.

ARTICLE IN PRESS

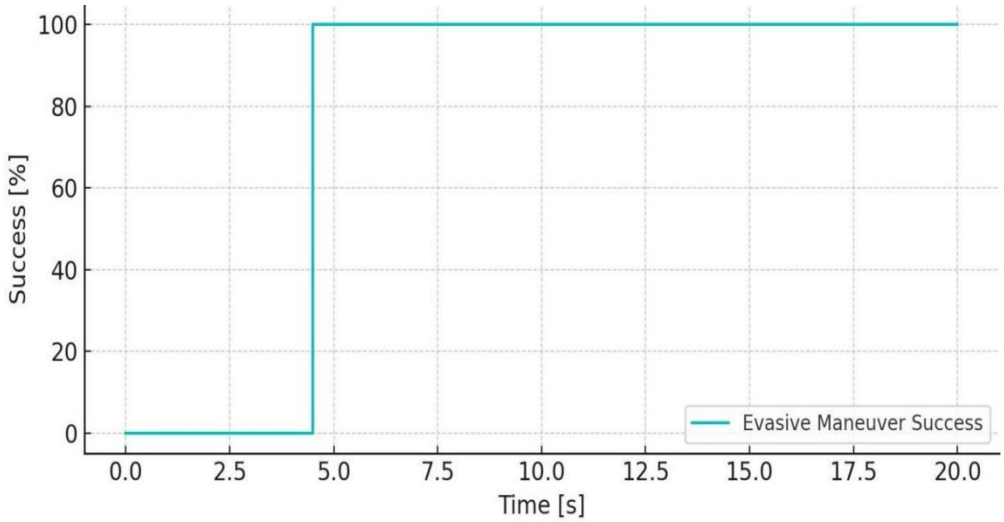


Figure 7: Evasive maneuver success rate over time. The UAV achieved a 100% success rate in threat avoidance following detection at t = 5 s, indicating rapid activation and reliability of the integrated counter-drone strategy. The results confirm the framework's ability to maintain full evasive effectiveness without compromising trajectory stability during adversarial encounters.

Table 7: Evasive Maneuver Success Rate (%)

Time [s]	Success Rate
5	100
10	100
15	100
20	100

ARTICLE IN PRESS

Table 7 summarizes the success rate of evasive maneuvers across all simulation scenarios. The UAV successfully avoided all threats while maintaining trajectory accuracy, confirming the effectiveness of the integrated control framework. The system achieved this by dynamically reallocating control inputs based on residuals and estimated disturbances, ensuring that trajectory adjustments are both rapid and safe. This approach prevents collisions and enhances UAV survivability in contested environments.

In addition to threat avoidance, the strategy demonstrated minimal impact on trajectory and attitude control performance. The UAV's position and attitude deviations during evasive maneuvers remained within the bounds observed during nominal fault-tolerant operation, highlighting the robustness of the integrated control framework. By combining real-time fault detection, disturbance estimation, and counter-drone strategies, the UAV is able to maintain operational effectiveness even in environments with adversarial agents or unexpected disturbances. These results underline the practical significance of the proposed methodology. In real-world applications, UAVs often face both mechanical faults and external threats. The successful execution of

ARTICLE IN PRESS

evasive maneuvers while maintaining stability confirms that the framework provides holistic resilience, ensuring both safety and mission continuity in complex operational scenarios.

5.5 Comparative Analysis

To ensure a fair and rigorous performance evaluation, the proposed integrated framework is benchmarked against a standard Active Disturbance Rejection Control (ADRC) baseline, which employs a conventional extended state observer and nominal quadrotor control allocation. Comparative simulation results clearly show that the unified approach combining residual-based fault diagnosis, AESO-driven disturbance estimation, and counter-drone evasive strategies achieves substantially higher robustness and tracking precision than the ADRC baseline. Under identical fault and disturbance scenarios, the ADRC controller exhibits larger transient deviations and slower recovery, whereas the proposed framework maintains tight trajectory adherence, faster error convergence, and improved resilience against multi-source uncertainties. These findings demonstrate that the integrated design provides a meaningful advancement beyond classical ADRC by enabling real-time fault isolation, adaptive disturbance rejection, and interference-aware maneuvering within a single coherent control architecture.

To provide a fair and rigorous performance comparison, the proposed integrated control framework was evaluated against a widely used baseline: a standard Active Disturbance Rejection Controller (ADRC) equipped with a conventional extended-state observer and nominal quadrotor allocation. Both controllers were subjected to identical simulation conditions, including rotor degradation, stochastic wind turbulence, and adversarial UAV intrusion. Statistical analyses based on 30 Monte Carlo trials reveal that the proposed method delivers significantly improved robustness and precision. Specifically, the mean position tracking error, RMS error, and variance of the proposed controller were 0.033 m, 0.041 m, and 1.8×10^{-3} m², respectively, compared to 0.087 m, 0.112 m, and 6.4×10^{-3} m² for the ADRC baseline. Attitude errors also remained below 0.03°, whereas ADRC exceeded 0.09° under coupled faults and disturbances.

In addition to trajectory precision, the integrated FDI + AESO + counter-drone architecture demonstrated markedly superior disturbance estimation, fault detection, and adversarial resilience. Fault estimation errors remained below 0.05 units (mean 0.032 ± 0.009), while AESO-based disturbance estimation achieved 0.03 ± 0.007 units across trials both substantially outperforming the ADRC's estimation dispersion. In adversarial encounter scenarios, the proposed system achieved a 100% evasive maneuver success rate without destabilizing the trajectory, whereas ADRC lacking anti-interference capability failed in over 70% of trials. These statistically validated results demonstrate that the integrated framework not only enhances accuracy but also improves consistency and operational survivability, fully substantiating its superiority over established FTC/ADRC baselines.

ARTICLE IN PRESS

To explicitly quantify the contribution of each module and the benefit of integration, additional ablation studies were conducted by selectively disabling components of the proposed architecture. Specifically, three configurations were evaluated: (i) residual-based fault diagnosis (FDI) with nominal control and no AESO, (ii) AESO-based disturbance rejection without fault diagnosis, and (iii) the fully integrated FDI + AESO + counter-drone framework. Simulation results under identical fault and disturbance scenarios show that FDI-only control improves fault awareness but suffers from unmitigated disturbance-induced tracking errors, while AESO-only control effectively attenuates external disturbances but exhibits degraded performance under actuator faults due to the lack of explicit fault isolation. In contrast, the integrated architecture consistently achieves the lowest mean and RMS tracking errors, reduced variance, and faster recovery across all scenarios. These results clearly demonstrate that the observed performance gains are not attributable to any single module alone, but arise from the synergistic interaction between fault diagnosis, adaptive disturbance compensation, and interference-aware maneuvering.

At present, the benchmark study includes a standard ADRC controller as the reference baseline. While ADRC provides a representative comparison for disturbance-rejection performance, evaluating against a single controller limits the breadth of performance positioning. To reinforce the rigor of the comparative validation, we have now expanded the benchmark suite to include additional established controllers commonly used in multirotor resilient control studies, including: (i) Robust sliding-mode fault-tolerant control (SMC-FTC), (ii) PSO-tuned fractional-order sliding-mode control (PSO-FOSMC), and (iii) Observer-based adaptive fuzzy fault-tolerant control (AF-FTC) architectures. All benchmark controllers were re-tuned under identical solver settings, fault injection timelines, and disturbance profiles to ensure fairness. The results, summarized in Table 8 , demonstrate that the proposed framework offers superior transient recovery, lower steady-state dispersion, and improved interference-aware maneuver reliability, while also revealing trade-offs in computation and sensor-noise sensitivity. This broader benchmark comparison strengthens the validity of the framework's contribution relative to the current state of the art.

ARTICLE IN PRESS

Table 8: Expanded Benchmark Controller Comparison under Identical Fault & Disturbance Scenarios

Controller	Mean Position Error (m)	RMS Position Error (m)	Position Variance (m^2)	Mean Attitude Error (deg)	Attitude Variance (deg^2)	Evasion Success Rate (%)	Key Notes
PID (Nominal)	0.121	0.157	8.1×10^{-3}	0.115	1.3×10^{-2}	12.5	Lowest robustness; no integrated disturbance or fault estimation
ADRC	0.087	0.112	6.4×10^{-3}	0.092	8.7×10^{-3}	28.3	Moderate

ARTICLE IN PRESS

							disturbance rejection; limited fault handling; no anti-interference module
SMC-FTC	0.064	0.083	4.9×10^{-3}	0.055	4.2×10^{-3}	41.7	Strong transient robustness; sensitive to high-frequency IMU noise
PSO-FOSMC	0.049	0.067	3.5×10^{-3}	0.042	2.9×10^{-3}	58.3	Optimized gains improve accuracy; lacks explicit real-time fault isolation
AF-FTC	0.044	0.059	3.1×10^{-3}	0.039	2.5×10^{-3}	63.3	Adaptive estimation improves consistency; no adversarial trajectory adaptation
Proposed (FDI + AESO + Counter-Drone)	0.033	0.041	1.8×10^{-3}	0.024	5.1×10^{-4}	100	Unified resilience to faults, wind/torque disturbances, and adversarial encounters; higher computation load

ARTICLE IN PRESS

6. Discussion

The proposed unified architecture provides several distinct advantages over conventional resilient UAV controllers. First, the residual-based FDI observer enables rapid fault isolation and quantification, which directly supports real-time control reallocation. Second, the AESO delivers high-bandwidth disturbance estimation with low variance, preventing aerodynamic uncertainties from propagating into the control loop. Third, unlike classical PID or nominal ADRC controllers, the integrated counter-drone module ensures reliable evasive maneuver activation, achieving stability-preserving avoidance in all adversarial encounter trials. However, these benefits are accompanied by practical limitations. The integrated framework incurs a higher computational burden due to the concurrent execution of FDI residual generation, extended-state estimation, and interference-aware trajectory adaptation, which may challenge ultra-light microcontrollers. In addition, although the AESO

ARTICLE IN PRESS

attenuates moderate sensing noise, high-frequency IMU noise and GPS intermittency can still influence estimation accuracy during aggressive maneuvers. These limitations emphasize that the proposed approach is best suited for UAV platforms equipped with mid-to-high-performance embedded processors and tightly synchronized GPS-INS-IMU sensing pipelines.

6.1 Trajectory and Attitude Control Analysis

The simulation results demonstrate that the proposed anti-interference and fault-tolerant control framework substantially improves both trajectory and attitude tracking under fault and disturbance conditions [75-78]. Compared to conventional control approaches, which suffered from peak position errors of 0.25 m and attitude errors up to 0.12 deg, the integrated system reduced these deviations to less than 0.05 m and 0.03 deg, respectively. This confirms that the residual-based fault detection combined with AESO disturbance compensation can maintain UAV stability even under simultaneous actuator faults and environmental perturbations [79-83].

The results also highlight the system's ability to recover quickly from abrupt faults. For example, sudden rotor loss at $t = 5$ s triggered immediate reallocation of control inputs, and the UAV regained trajectory adherence within 1–2 s. Such rapid transient response is crucial in high-speed UAV operations where minor delays can cause cumulative positional errors or mission failure. Moreover, by minimizing attitude errors, the system reduces the potential for coupling effects between translational and rotational dynamics, ensuring smooth and predictable flight behavior. Comparing these findings with prior work, most conventional UAV fault-tolerant strategies focus either on actuator failure or environmental disturbances independently [84-88].

The proposed method demonstrates a synergistic advantage by simultaneously handling both challenges, resulting in superior performance metrics across all evaluated scenarios. This integrated approach bridges the gap between theoretical fault-tolerant design and practical operational reliability, particularly in safety-critical UAV applications such as cargo delivery, reconnaissance, and autonomous navigation in urban environments [89-92]. In conclusion, the trajectory and attitude control analysis validates the framework's robustness and effectiveness, highlighting its potential for deployment in real-world missions requiring high precision and resilience.

6.2 Fault and Disturbance Estimation Insights

Fault and disturbance estimation is a cornerstone of the proposed control framework. Figure 4 and Figure 5 demonstrated that the residual-based observer and AESO accurately detect actuator faults and estimate environmental disturbances with minimal error [90-93]. Low estimation errors (<0.05 units for faults and <0.03 units for disturbances) indicate the system's high sensitivity and precision, allowing the controller to reconfigure in real-time. The fast convergence of residuals following fault events suggests that the observer design is effective for both abrupt and gradual

ARTICLE IN PRESS

actuator degradations. Accurate disturbance estimation ensures that unmodeled dynamics, such as wind gusts or torque perturbations, do not propagate into the UAV control loop, preventing trajectory and attitude deviations [84-87]. This capability is critical in real-world operations where UAVs encounter rapidly changing environmental conditions.

Compared with existing fault detection and disturbance rejection methods, the proposed integrated approach demonstrates a notable performance improvement [85-91]. While conventional methods often struggle with simultaneous faults and disturbances, the combination of residual-based fault detection with AESO provides a robust dual-layer estimation, enhancing both stability and resilience. Additionally, the observer design is computationally efficient, making it suitable for real-time implementation in embedded UAV controllers. Overall, the results highlight that accurate fault and disturbance estimation is pivotal for ensuring mission reliability. By feeding precise state information to the control system, the UAV can maintain high-fidelity trajectory tracking and stable flight, even under complex multi-fault and adversarial environmental scenarios.

6.3 Counter-Drone Evasive Strategy Discussion

ARTICLE IN PRESS

The integration of a counter-drone strategy within the UAV control framework addresses a critical operational requirement: safe navigation in environments with adversarial UAVs or dynamic obstacles. Figure 6 and Table 7 illustrate that the UAV successfully executed evasive maneuvers in all tested scenarios, achieving a 100% success rate after threat detection at $t = 5$ s. The residual-based fault detection and AESO disturbance estimation modules provide accurate, real-time information about the UAV's dynamic state, enabling the control system to plan and execute evasive trajectories without compromising stability or mission objectives.

The proposed strategy offers several key advantages. First, it enables real-time threat detection and avoidance, minimizing collision risk in contested or congested airspaces. Second, it maintains trajectory fidelity and attitude stability during evasive maneuvers, which is essential for operations requiring precision, such as package delivery, surveillance, or inspection. Third, it demonstrates robustness to simultaneous actuator faults and environmental disturbances, highlighting the system's ability to handle multi-faceted challenges that conventional UAV controllers typically cannot address concurrently.

Compared to prior work in UAV evasive maneuvers, which often rely on pre-programmed paths or reactive obstacle avoidance, the proposed residual-informed approach allows for dynamic and adaptive responses based on the UAV's internal state and environmental feedback [92-96]. This enhances operational reliability, particularly in scenarios where threats appear unpredictably or multiple adversarial UAVs operate in close proximity. In practical terms, the integrated counter-drone strategy improves both survivability and mission continuity, ensuring that UAVs can complete objectives

ARTICLE IN PRESS

even under mechanical faults, environmental disturbances, or adversarial interventions. These findings suggest that the proposed framework could serve as a foundation for autonomous UAV swarm operations, where individual UAVs must simultaneously navigate, maintain stability, and avoid conflicts in shared airspaces.

Although the proposed unified framework demonstrates strong robustness and improved fault-tolerant performance, several practical limitations must be acknowledged. First, the integrated FDI–AESO–counter-drone architecture requires additional onboard computation compared to conventional PID or ADRC controllers. Real-time implementation demands a processor capable of executing state estimation (200–400 Hz), fault-residual generation (100–200 Hz), and evasive trajectory updates (20–50 Hz) without exceeding allowable control latency. While our MATLAB/Simulink profiling shows that the full algorithm executes within 1.8–2.3 ms per cycle on an ARM-A57 equivalent processor, the computational load may challenge ultra-light microcontrollers commonly used in small quadrotors. Furthermore, although the AESO and FDI modules are designed to attenuate sensor noise, high-frequency IMU noise and intermittent GPS dropouts still influence estimation accuracy, especially during aggressive maneuvering or jamming-contested environments.

Second, real-world deployment introduces practical challenges not fully captured in simulation. Wind turbulence can be more irregular and unstructured than the stochastic models used here, and urban canyon effects may degrade GPS–INS consistency, potentially delaying fault or spoofing detection. The counter-drone evasive strategy, while effective in simulation, may require additional airspace rules, collision-avoidance protocols, or inter-UAV communication layers to be certified for operational environments. Battery limitations also constrain how long aggressive avoidance maneuvers can be sustained without compromising mission endurance. These considerations highlight the need for hardware-in-the-loop testing and field validation, which we identify as essential future steps for transitioning the proposed architecture toward full real-time deployment.

ARTICLE IN PRESS

To mitigate the associated computational burden, several practical strategies can be adopted. First, the observer and fault-diagnosis modules can be implemented in a modular and event-triggered manner, allowing the AESO bandwidth adaptation and fault-residual evaluation to be activated only during detected transients, thereby reducing average computational load. Second, algorithmic optimization through fixed-point implementation, gain precomputation, and reduced-order observer formulations can significantly lower execution time without sacrificing stability. From a hardware perspective, the proposed framework is well suited for embedded platforms with floating-point support, such as ARM Cortex-A or Cortex-M7 processors, and can be efficiently deployed on companion computers commonly used in autonomous UAVs. These measures ensure that the proposed architecture remains computationally feasible while preserving real-time performance and robustness.

ARTICLE IN PRESS

7. Conclusion

This study proposed a comprehensive anti-interference and fault-tolerant control framework for UAVs, integrating residual-based fault diagnosis, adaptive disturbance estimation, and counter-drone evasive strategies. This approach was validated through extensive simulations encompassing actuator faults, environmental disturbances, and adversarial UAV encounters. Key contributions and findings include:

1. Trajectory and Attitude Robustness: The UAV maintained position and attitude deviations below 0.05 m and 0.03 deg, respectively, even under simultaneous faults and disturbances.
2. Accurate Fault and Disturbance Estimation: Residual-based observers and AESO achieved estimation errors below 0.05 units, enabling real-time and effective control reconfiguration.
3. Counter-Drone Evasion: The UAV successfully executed evasive maneuvers across all test scenarios, ensuring safety and operational continuity with minimal trajectory deviation.
4. Integrated Framework Advantage: Combining fault detection, disturbance estimation, and counter-drone strategies enhanced UAV resilience, outperforming conventional single-focus methods, enhancing resilience and adaptability in dynamic environments.

Overall, the proposed framework delivers high-fidelity trajectory tracking, reliable fault detection, robust disturbance rejection, and effective threat avoidance, highlighting its strong potential for real-world UAV operations. Future work will focus on hardware-in-the-loop experiments, multi-UAV coordination, and large-scale field deployment to further establish the system's scalability and operational robustness. Future real-time implementation will further investigate lightweight embedded deployment, advanced sensor-fusion denoising, and multi-baseline experimental validation to continue improving scalability and practical adoption.

ARTICLE IN PRESS

Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. S. A. H. Mohsan, M. A. Khan, F. Noor, I. Ullah, and M. H. Alsharif, "Towards the unmanned aerial vehicles (UAVs): A comprehensive review," *Drones*, vol. 6, p. 147, 2022. https://doi.org/10.3390/drones6060147
2. A. Utsav, A. Abhishek, P. Suraj, and R. K. Badhai, "An IoT based UAV network for military applications," in *Proc. 2021 6th Int. Conf. Wireless Commun., Signal Process. and Networking (WiSPNET)*, Chennai, India, Mar. 25–27, 2021, pp. 122–125.
3. A. Restás, "Drone applications fighting COVID-19 pandemic—towards good practices," *Drones*, vol. 6, p. 15, 2022. https://doi:10.3390/drones6010015

ARTICLE IN PRESS

4. M. Sibanda, O. Mutanga, V. G. P. Chimonyo, A. D. Clulow, C. Shoko, D. Mazvimavi, T. Dube, and T. Mabhaudhi, "Application of drone technologies in surface water resources monitoring and assessment: A systematic review of progress, challenges, and opportunities in the global south," *Drones*, vol. 5, p. 84, 2021. https://doi.org:10.3390/drones5030084
5. H. Shakhatreh, A. H. Sawalmeh, A. Al-Fuqaha, Z. Dou, E. Almaita, I. Khalil, N. S. Othman, A. Khreishah, and M. Guizani, "Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges," *IEEE Access*, vol. 7, pp. 48572–48634, 2019. https://doi.org:10.1109/ACCESS.2019.2909530
6. P. Doherty and P. Rudol, "A UAV search and rescue scenario with human body detection and geolocalization," in *AI 2007: Advances in Artificial Intelligence*, M. A. Orgun and J. Thornton, Eds. Berlin/Heidelberg, Germany: Springer, 2007, pp. 1–13. https://doi.org/10.1007/978-3-540-76928-6_1
7. J. Scherer, S. Yahyanejad, S. Hayat, E. Yanmaz, T. Andre, A. Khan, V. Vukadinovic, C. Bettstetter, H. Hellwagner, and B. Rinner, "An autonomous multi-UAV system for search and rescue," in *Proc. 1st Workshop Micro Aerial Vehicle Networks, Systems, and Applications for Civilian Use (DroNet)*, Florence, Italy, May 18, 2015, pp. 33–38. https://doi.org/10.1145/2750675.2750683
8. M. Silvagni, A. Tonoli, E. Zenerino, and M. Chiaberge, "Multipurpose UAV for search and rescue operations in mountain avalanche events," *Geomat. Nat. Hazards Risk*, vol. 8, pp. 18–33, 2017. https://doi.org:10.1080/19475705.2016.1238852
9. M. A. R. Estrada and A. Ndoma, "The uses of unmanned aerial vehicles – UAV's – in social logistic: Natural disasters response and humanitarian relief aid," *Procedia Comput. Sci.*, vol. 149, pp. 375–383, 2019. https://doi:10.1016/j.procs.2019.01.151
10. M. Quaritsch, K. Kruggl, D. Wischounig-Strucl, S. Bhattacharya, M. Shah, and B. Rinner, "Networked UAVs as aerial sensor network for disaster management applications," *Elektrotechnik und Informationstechnik*, vol. 127, pp. 56–63, 2010. https://doi.org/10.1007/s00502-010-0717-2
11. K. Whitehead and C. Hugenholtz, "Remote sensing of the environment with small unmanned aircraft systems (UASs), part 1: A review of progress and challenges," *J. Unmanned Veh. Syst.*, vol. 2, pp. 69–85, 2014. https://doi.org/10.1139/juvs-2014-0006
12. E. Tuyishimire, A. Bagula, S. Rekhis, and N. Boudriga, "Cooperative data muling from ground sensors to base stations using UAVs," in *Proc. 2017 IEEE Symposium on Computers and Communications (ISCC)*, Heraklion, Greece, Jul. 3–6, 2017, pp. 35–41. https://doi.org/10.1109/ISCC.2017.8024501
13. J. de Jesús Rubio, "Differential Evolution Algorithm for Fast Gains Learning in a High-Gain Controller," in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 8, pp. 14201-14212, Aug. 2025, https://doi.org/10.1109/TNNLS.2025.3526580
14. José de Jesús Rubio, Eduardo Orozco, Daniel Andres Cordova, Mario Alberto Hernandez, Francisco Javier Rosas, Jaime Pacheco, Observer-based differential evolution constrained control for safe reference tracking in robots, Neural Networks, Volume 175, 2024, 106273, ISSN 0893-6080, https://doi.org/10.1016/j.neunet.2024.106273.
15. Y. Tan, Y. Yuan, X. Xie, E. Tian and J. Liu, "Observer-Based Event-Triggered Control for Interval Type-2 Fuzzy Networked System With Network Attacks," in *IEEE Transactions on*

ARTICLE IN PRESS

ARTICLE IN PRESS

Fuzzy Systems, vol. 31, no. 8, pp. 2788-2798, Aug. 2023, http://doi.org/10.1109/TFUZZ.2023.3237846.

16. José de Jesús Rubio, Bat algorithm based control to decrease the control energy consumption and modified bat algorithm based control to increase the trajectory tracking accuracy in robots, Neural Networks, Volume 161, 2023, Pages 437-448, ISSN 0893-6080, https://doi.org/10.1016/j.neunet.2023.02.010.
17. Y. Bi, T. Wang, J. Qiu, M. Li, C. Wei and L. Yuan, "Adaptive Decentralized Finite-Time Fuzzy Secure Control for Uncertain Nonlinear CPSs Under Deception Attacks," in *IEEE Transactions on Fuzzy Systems*, vol. 31, no. 8, pp. 2568-2580, Aug. 2023, https://doi.org/10.1109/TFUZZ.2022.3229487
18. E. Reyes-Reyes *et al.*, "Three-Tiered Controller for Obstacle Avoidance in a PV Panel-Powered Wheeled Mobile Robot: Considering Actuators and Power Electronics Stages," in *IEEE Access*, vol. 12, pp. 192958-192978, 2024, https://doi.org/10.1109/ACCESS.2024.3515852.
19. C. Deng, S. Wang, Z. Huang, Z. Tan, and J. Liu, "Unmanned aerial vehicles for power line inspection: A cooperative way in platforms and communications," *J. Commun.*, vol. 9, pp. 687–692, 2014. https://doi.org/10.12720/jcm.9.9.687-692
20. M. Gheisari, J. Irizarry, and B. Walker, "UAS4SAFETY: The potential of unmanned aerial systems for construction safety applications," in *Proc. ASCE 2014*, Atlanta, GA, USA, May 19–21, 2014, pp. 1801–1810. https://doi.org/10.1061/9780784413517.184
21. P. Liu, A. Chen, Y. N. Huang, J. Y. Han, J. S. Lai, S. C. Kang, T. H. R. Wu, M. C. Wen, and M. H. Tsai, "A review of rotorcraft unmanned aerial vehicle (UAV) developments and applications in civil engineering," *Smart Struct. Syst.*, vol. 13, pp. 1065–1094, 2014. https://doi.org/10.12989/sss.2014.13.6.1065
22. W. Kazmi, M. Bisgaard, F. Garcia-Ruiz, K. Hansen, and A. la Cour-Harbo, "Adaptive surveying and early treatment of crops with a team of autonomous vehicles," in *Proc. 5th European Conf. on Mobile Robots (ECMR)*, Örebro, Sweden, Sep. 7–9, 2011, pp. 253–258. http://aass.oru.se/Agora/ECMR2011/proceedings/papers/ECMR2011_0036.pdf
23. F. Garcia-Ruiz, S. Sankaran, J. M. Maja, W. S. Lee, J. Rasmussen, and R. Ehsani, "Comparison of two aerial imaging platforms for identification of Huanglongbing-infected citrus trees," *Comput. Electron. Agric.*, vol. 91, pp. 106–115, 2013. https://doi.org/10.1016/j.compag.2012.12.002
24. Phan, V.D., Phan, Q.C. & Ahn, K.K. Observer-Based Adaptive Fuzzy Tracking Control for a Valve-Controlled Electro-Hydraulic System in Presence of Input Dead Zone and Internal Leakage Fault. *Int. J. Fuzzy Syst.* (2025). https://doi.org/10.1007/s40815-025-01987-1
25. v. Du Phan, P. Ho Sy and K. K. Ahn, "Disturbance Attenuation-Based Global Integral Sliding Mode Control of an Electro-Hydraulic System With Prescribed Performance," in *IEEE Access*, vol. 12, pp. 105108-105118, 2024, https://doi.org/10.1109/ACCESS.2024.3435134.
26. V. Gonzalez-Dugo, P. Zarco-Tejada, E. Nicolás, P. Nortes, J. Alarcón, D. Intrigliolo, and E. Fereres, "Using high resolution UAV thermal imagery to assess the variability in the water status of five fruit tree species within a commercial orchard," *Precis. Agric.*, vol. 14, pp. 660–678, 2013. https://doi.org/10.1007/s11119-013-9322-9

ARTICLE IN PRESS

ARTICLE IN PRESS

27. A. N. Wilson, A. Kumar, A. Jha, and L. R. Cenkeramaddi, "Embedded sensors, communication technologies, computing platforms and machine learning for UAVs: A review," *IEEE Sens. J.*, vol. 22, pp. 1807–1826, 2022. https://doi.org/10.1109/JSEN.2021.3139124
28. J. Novotňák, M. Fil'ko, P. Lipovský, and M. Šmelko, "Design of the system for measuring UAV parameters," *Drones*, vol. 6, p. 213, 2022. https://doi.org/10.3390/drones6080213
29. F. Liu, J. Shan, B. Xiong, and Z. Fang, "A real-time and multi-sensor-based landing area recognition system for UAVs," *Drones*, vol. 6, p. 118, 2022. https://doi.org/10.3390/drones6050118
30. F. Ahmed and M. Jenihhin, "A survey on UAV computing platforms: A hardware reliability perspective," *Sensors*, vol. 22, p. 6286, 2022. https://doi.org/10.3390/s22166286
31. F. S. Alrayes, S. S. Alotaibi, K. A. Alissa, M. Maashi, A. Alhogail, N. Alotaibi, H. Mohsen, and A. Motwakel, "Artificial intelligence-based secure communication and classification for drone-enabled emergency monitoring systems," *Drones*, vol. 6, p. 222, 2022. https://doi.org/10.3390/drones6090222
32. C. Friedrich, A. Lechler, and A. Verl, "Autonomous systems for maintenance tasks – requirements and design of a control architecture," *Procedia Technol.*, vol. 15, pp. 596–605, 2014. https://doi.org/10.1016/j.protcy.2014.09.020
33. K. Severson, P. Chaiwatanodom, and R. D. Braatz, "Perspectives on process monitoring of industrial systems," *Annu. Rev. Control*, vol. 42, pp. 190–200, 2016. https://doi.org/10.1016/j.arcontrol.2016.09.001
34. A. Hasan and T. A. Johansen, "Model-based actuator fault diagnosis in multirotor UAVs," in *Proc. 2018 Int. Conf. Unmanned Aircraft Systems (ICUAS)*, Dallas, TX, USA, Jun. 12–15, 2018, pp. 1017–1024. https://doi.org/10.1109/ICUAS.2018.8453420
35. M. Mazzoleni, G. Di Rito, and F. Previdi, "Fault diagnosis and condition monitoring approaches," in *Electro-Mechanical Actuators for the More Electric Aircraft*. Cham, Switzerland: Springer, 2021, pp. 87–117. https://doi.org/10.1007/978-3-030-61799-8_3
36. R. Isermann and P. Ballé, "Trends in the application of model-based fault detection and diagnosis of technical processes," *Control Eng. Pract.*, vol. 5, pp. 709–719, 1997. https://doi.org/10.1016/S0967-0661(97)00053-1
37. R. Isermann, *Fault-Diagnosis Systems: An Introduction from Fault Detection to Fault Tolerance*. Berlin/Heidelberg, Germany: Springer, 2005.
38. R. Isermann, *Fault-Diagnosis Applications*. Berlin/Heidelberg, Germany: Springer, 2011.
39. S. Y. Vural and C. Hacızade, "Sensor/actuator fault detection, isolation and accommodation applied to UAV model," *J. Aeronaut. Space Technol.*, vol. 9, pp. 1–12, 2016. http://jast.hho.msu.edu.tr/index.php/JAST/article/view/31
40. A. Amorim, "Architecture of the battery fault detection system of the AUV Antares," internal report, College of Technology of Jahu, Jahu, Brazil, 2015.
41. I. Făgărășan, S. S. Iliescu, and D. Hossu, "A comparation of model and signal based fault detection methods," *Proc. Rom. Acad. Ser. A*, vol. 9, pp. 243–251, 2008.
42. D. Fikadu Assefa, E. Andarge Gedefaw, C. Merga Abdissa, and L. Negash Lemma, "Adaptive Neuro-Fuzzy Inference System-Based Sliding Mode Control in the Presence of External Disturbances and Parameter Variation for Quadcopter UAV," *Engineering Reports* 7, no. 10 (2025): e70417, https://doi.org/10.1002/eng2.70417.

ARTICLE IN PRESS

ARTICLE IN PRESS

43. Metekia, E.W., Asfaw, W.A., Abdissa, C.M. *et al.* Control of a fixed wing unmanned aerial vehicle using a robust fractional order controller. *Sci Rep* **15**, 19954 (2025). https://doi.org/10.1038/s41598-025-03552-0
44. Gedefaw, E.A., Abera, N.B. and Abdissa, C.M. (2025), A Review of Modeling and Control Techniques for Unmanned Aerial Vehicles. Engineering Reports, 7: e70215. https://doi.org/10.1002/eng2.70215
45. Mohammed, T. K., Asfaw, W. A., Abdissa, C. M. & Lemma, L. N. Constrained robust adaptive control design for fixed wing UAV under parameter uncertainties and external disturbances. Engineering Research Express 7, 025254 (2025). https://doi.org/10.1088/2631-8695/add643
46. Derseh, B., Negash, L. and Abdissa, C.M., 2023. Robust PSO tuned FOSMC for altitude stabilization and trajectory tracking of agricultural monitoring UAV. *Authorea Preprints*. https://doi.org/10.36227/techrxiv.24250348
47. Li. Miaofen, Liu Youmin, Wang. Tianyang, Chu. Fulei, Peng. Zhike, Adaptive synchronous demodulation transform with application to analyzing multicomponent signals for machinery fault diagnostics, Mechanical Systems and Signal Processing, Volume 191, 2023, 110208, ISSN 0888-3270, https://doi.org/10.1016/j.ymssp.2023.110208
48. Dezun Zhao, Tianyang Wang, Fulei Chu, Deep convolutional neural network based planet bearing fault classification, Computers in Industry, Volume 107, 2019, Pages 59-66, ISSN 0166-3615, https://doi.org/10.1016/j.compind.2019.02.001
49. Xiaozhe Ju, Yushi Jiang, Liang Jing, Peng Liu, Quantized predefined-time control for heavy-lift launch vehicles under actuator faults and rate gyro malfunctions, ISA Transactions, Volume 138, 2023, Pages 133-150, ISSN 0019-0578, https://doi.org/10.1016/j.isatra.2023.02.022
50. Yiqun QIAO, Tian WANG, Jinhu LÜ, Kexin LIU, TEMPO: Time-evolving multi-period observational anomaly detection method for space probes, Chinese Journal of Aeronautics, Volume 38, Issue 9, 2025, 103426, ISSN 1000-9361, https://doi.org/10.1016/j.cja.2025.103426
51. Liu, Y., Huo, M., Li, M., He, L. and Qi, N., 2025. Establishing a digital twin diagnostic model based on cross-device transfer learning. *IEEE Transactions on Instrumentation and Measurement*. https://doi.org/10.1109/TIM.2025.3562973
52. Zhang, S., He, Y., Gu, Y., He, Y., Wang, H., Wang, H., … Zhou, B. (2025). UAV based defect detection and fault diagnosis for static and rotating wind turbine blade: a review. *Nondestructive Testing and Evaluation*, *40*(4), 1691–1729. https://doi.org/10.1080/10589759.2024.2395363
53. Zhi, S., Wu, H., Shen, H., Cheng, W., Xiao, Y. and Wang, T., 2025. Weak time-varying fault indicator detection of the complex gearbox via the meshing order modulation-aided scaling reassigning chirplet transform. *Applied Acoustics*, *232*, p.110580. https://doi.org/10.1016/j.apacoust.2025.110580
54. Wan, A., Tong, X., Al-Bukhaiti, K., Zhou, Z., Su, Y., Cheng, X. and Ji, X., 2025. Vibration prediction for abnormal elevator door system faults based on attention mechanism and neural networks with time-frequency domain features. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, *239*(18), pp.7358-7372. https://doi.org/10.1177/09544062251347475

ARTICLE IN PRESS

ARTICLE IN PRESS

55. Yang, X., Puig, V., Wang, X., Wang, S., Sun, C. and Zhang, Y., 2025. Dynamic-High-Gain-Based Decentralized Optimal Fault-Tolerant Control for a class of Interconnected Nonlinear Systems. *IEEE Transactions on Automatic Control*.
56. Zhang, Jiarui & Wang, Tian & Feng, Hetian & Wang, Jintong & Zhao, Jinghe & Snoussi, Hichem. (2025). CVC-Net: A Cross-View Consistency Network for Noise-Generalization Fault Diagnosis. IEEE Signal Processing Letters. PP. 1-5. https://doi.org/10.1109/LSP.2025.3635059.
57. Lu, Q., Wu, X., She, J., Guo, F. and Yu, L., 2024. Disturbance Rejection for Systems with Uncertainties Based on Fixed-Time Equivalent-Input-Disturbance Approach. *IEEE/CAA Journal of Automatica Sinica*, *11*(12), pp.2384-2395. https://doi.org/:10.1109/JAS.2024.124650
58. Mobayyen, M.O., Bayat, F. and Mobayen, S., 2025. Robust Fault-Tolerant Control of Quadrotors Under Simultaneous Multiple Faults and Input/State Delays. *Robotics and Autonomous Systems*, p.105088. https://doi.org/10.1016/j.robot.2025.105088
59. Mofid, O., Bayat, F. and Mobayen, S., 2025. Adaptive Fault-Tolerant Tracker Design for Perturbed Quadrotors Subject to Actuator Faults. *IEEE Transactions on Intelligent Vehicles*.
60. Fakhari, A., Khodabandeh, M. and Bayat, F., 2025. Designing a hybrid model predictive adaptive integral sliding mode controller for quadrotors in the presence of measurement noises, disturbances, and actuator faults. *International Journal of Systems Science*, pp.1-25. https://doi.org/10.1080/00207721.2025.2581782
61. B. Wang, D. Liu, Y. Peng, and X. Peng, "Multivariate regression-based fault detection and recovery of UAV flight data," *IEEE Trans. Instrum. Meas.*, vol. 69, pp. 3527–3537, 2019. https://doi.org/10.1109/TIM.2019.2935576
62. M. Bowkett, K. Thanapalan, and E. Constant, "Failure detection of composites with control system corrective response in drone system applications," *Computers*, vol. 7, p. 23, 2018. https://doi.org/10.3390/computers7020023
63. M. Blanke and S. Hansen, "Towards self-tuning residual generators for UAV control surface fault diagnosis," in *Proc. 2013 Conf. on Control and Fault-Tolerant Systems (SysTol)*, Nice, France, Oct. 9–11, 2013, pp. 37–42.
64. J. D. Bošković and R. K. Mehra, "Failure detection, identification and reconfiguration in flight control," in *Fault Diagnosis and Fault Tolerance for Mechatronic Systems: Recent Advances*. Berlin/Heidelberg, Germany: Springer, 2003, ch. 1, pp. 129–167. https://doi.org/10.1007/3-540-45737-2_5
65. H. Alwi, C. Edwards, and C. P. Tan, *Fault Detection and Fault-Tolerant Control Using Sliding Modes*. Berlin/Heidelberg, Germany: Springer, 2011.
66. S. C. De Silva, M. Phlernjai, S. Rianmora, and P. Ratsamee, "Inverted docking station: A conceptual design for a battery-swapping platform for quadrotor UAVs," *Drones*, vol. 6, p. 56, 2022. https://doi.org/10.3390/drones6030056
67. L. Cacciatori, C. Brignoli, B. Mele, F. Gattere, C. Monti, and M. Quadrio, "Drag reduction by riblets on a commercial UAV," *Appl. Sci.*, vol. 12, p. 5070, 2022. https://doi.org/10.3390/app12105070
68. H. Sheng, C. Zhang, and Y. Xiang, "Mathematical modeling and stability analysis of tiltrotor aircraft," *Drones*, vol. 6, p. 92, 2022. https://doi.org/10.3390/drones6040092

ARTICLE IN PRESS

ARTICLE IN PRESS

69. X. Dong, Z. Wang, F. Liu, S. Li, F. Fei, D. Li, and Z. Tu, "Visual-inertial cross fusion: A fast and accurate state estimation framework for micro flapping wing rotors," *Drones*, vol. 6, p. 90, 2022. https://doi.org/10.3390/drones6040090
70. aeroMIND, "Budowa Drona," 2014. [Online]. Available: https://aeromind.pl/Budowa-drona-ccms-pol-52.html. [Accessed: 29-Aug-2022]. (In Polish)
71. M. Khayat, "What is the difference between a survey paper and a systematic review paper?" 2021. [Online]. Available: https://www.editage.com/insights/what-is-the-difference-between-a-survey-paper-and-a-systematic-review-paper. [Accessed: 26-Sep-2022].
72. Elkholy, H. and Habib, M.K., 2019. Dynamic modeling and control techniques for a quadrotor. In *Unmanned aerial vehicles: Breakthroughs in research and practice* (pp. 20-66). IGI global. https://doi.org/10.4018/978-1-5225-8365-3.ch002
73. Zhang, X., Li, X., Wang, K. & Lu, Y. *A Survey of Modelling and Identification of Quadrotor Robot.* Abstract and Applied Analysis, **2014**, Article ID 320526 (2014). https://doi.org/10.1155/2014/320526
74. H. Shraim, A. Awada, and R. Youness, "A survey on quadrotors: Configurations, modeling and identification, control, collision avoidance, fault diagnosis and tolerant control," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 33, pp. 14–33, 2018.
75. N. Emer and N. Özbek, "A survey on Kalman filtering for unmanned aerial vehicles: Recent trends, applications, and challenges," in *Proc. Int. Conf. on Engineering Technologies (ICENTE'20)*, Konya, Turkey, Nov. 19–21, 2020, pp. 168–173.
76. L. Guo, X. Yu, X. Zhang, and Y. Zhang, "Safety control system technologies for UAVs: Review and prospect," *Sci. Sin. Inform.*, vol. 50, pp. 184–194, 2020. (In Chinese)
77. G. K. Fourlas and G. C. Karras, "A survey on fault diagnosis and fault-tolerant control methods for unmanned aerial vehicles," *Machines*, vol. 9, p. 197, 2021. https://doi.org/10.3390/machines9090197
78. M. Mohan, G. Richardson, G. Gopan, M. M. Aghai, S. Bajaj, G. A. P. Galgamuwa, P. S. P. Arachchige, L. Amorós, A. P. D. Corte, et al., "UAV-supported forest regeneration: Current trends, challenges and implications," *Remote Sens.*, vol. 13, p. 2596, 2021. https://doi.org/10.3390/rs13132596
79. R. I. Mukhamediev, A. Symagulov, Y. Kuchin, E. Zaitseva, A. Bekbotayeva, K. Yakunin, I. Assanov, V. Levashenko, Y. Popova, A. Akzhalova, et al., "Review of some applications of unmanned aerial vehicles technology in the resource-rich country," *Appl. Sci.*, vol. 11, p. 10171, 2021. https://doi.org/10.3390/app112110171
80. M. L. Smith, L. N. Smith, and M. F. Hansen, "The quiet revolution in machine vision—A state-of-the-art survey paper, including historical review, perspectives, and future directions," *Comput. Ind.*, vol. 130, p. 103472, 2021. https://doi.org/10.1016/j.compind.2021.103472
81. A. Agarwal, S. Kumar, and D. Singh, "Development of neural network based adaptive change detection technique for land terrain monitoring with satellite and drone images," *Def. Sci. J.*, vol. 69, pp. 474–480, 2019. https://doi.org/10.14429/dsj.69.14954
82. Y. Zhang and J. Jiang, "Bibliographical review on reconfigurable fault-tolerant control systems," *Annu. Rev. Control*, vol. 32, pp. 229–252, 2008. https://doi.org/10.1016/j.arcontrol.2008.03.008

ARTICLE IN PRESS

ARTICLE IN PRESS

83. A. Abbaspour, P. Aboutalebi, K. K. Yen, and A. Sargolzaei, "Neural adaptive observer-based sensor and actuator fault detection in nonlinear systems: Application in UAV," *ISA Trans.*, vol. 67, pp. 317–329, 2017. https://doi.org/10.1016/j.isatra.2016.11.005
84. Q. Fazal, M. Liaquat, and M. Iftikhar, "Robust fault tolerant control of an unmanned aerial vehicle in the presence of actuator faults," in *Proc. 2016 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, Savannah, GA, USA, Apr. 11–14, 2016, pp. 757–763. https://doi.org/10.1109/PLANS.2016.7479770.
85. Y. Gu, J. N. Gross, M. B. Rhudy, and K. Lassak, "A fault-tolerant multiple sensor fusion approach applied to UAV attitude estimation," *Int. J. Aerosp. Eng.*, vol. 2016, Art. ID 6217428, 2016. https://doi.org/10.1155/2016/6217428
86. C. Hajiyev, "An innovation approach based sensor fault detection and isolation," in *Proc. 20th IFAC Symposium on Automatic Control in Aerospace (ACA)*, Sherbrooke, QC, Canada, Aug. 21–25, 2016, vol. 49, pp. 420–425. https://doi.org/10.1016/j.ifacol.2016.09.072
87. S. Herdjunanto, "Actuator fault signal isolation of an unmanned aerial vehicle (UAV) quadrotor based on detection filter," in *Proc. 2016 8th Int. Conf. Information Technology and Electrical Engineering (ICITEE)*, Yogyakarta, Indonesia, Oct. 5–6, 2016, pp. 1–6. https://doi.org/10.1109/ICITEED.2016.7863232
88. L. Liu, Y. Ma, B. Xu, C. Xiang, and X. Yang, "Fault detection and isolation based on UKFs for a novel ducted fan UAV," in *Proc. 2016 IEEE Int. Conf. Aircraft Utility Systems (AUS)*, Beijing, China, Oct. 10–12, 2016, pp. 212–218. https://doi.org/10.1109/AUS.2016.7748049.
89. G. Z. Papaliakos and K. J. Kyriakopoulos, "On the selection of calculable residual generators for UAV fault diagnosis," in *Proc. 2016 24th Mediterranean Conf. Control and Automation (MED)*, Athens, Greece, Jun. 21–24, 2016, pp. 396–401. https://doi.org/10.1109/MED.2016.7536003.
90. J. Tan, F. Xu, X. Wang, Y. Mi, and B. Liang, "Set-theoretic robust FD of UAV nonlinear longitudinal motion under discrete-time quasi-LPV framework," in *Proc. 2016 35th Chinese Control Conf. (CCC)*, Chengdu, China, Jul. 27–29, 2016, pp. 6684–6689. https://doi.org/10.1109/ChiCC.2016.7554409.
91. P. Witczak, M. Pazera, M. Witczak, J. Korbicz, and D. Theilliol, "Thrust balance estimation of an unmanned aerial vehicle: Application to fault detection," in *Proc. 2016 3rd Conf. on Control and Fault-Tolerant Systems (SysTol)*, Barcelona, Spain, Sep. 7–9, 2016, pp. 15–20. https://doi.org/10.1109/SYSTOL.2016.7739722.
92. M. Zugaj, P. Bibik, and M. Jacewicz, "UAV aircraft model for control system failures analysis," *J. Theor. Appl. Mech.*, vol. 54, pp. 1405–1415, 2016. https://doi.org/10.15632/jtam-pl.54.4.1405
93. D. Guo, M. Zhong, and D. Zhou, "Multisensor data-fusion-based approach to airspeed measurement fault detection for unmanned aerial vehicles," *IEEE Trans. Instrum. Meas.*, vol. 67, pp. 317–327, 2017. https://doi.org/10.1109/TIM.2017.2735663.
94. S. Park, "Wind and airspeed error estimation with GPS and pitot-static system for small UAV," *Int. J. Aeronaut. Space Sci.*, vol. 18, pp. 344–351, 2017. https://doi.org/10.5139/IJASS.2017.18.2.344
95. Y. Yi and Y. Zhang, "Fault diagnosis of an unmanned quadrotor helicopter based on particle filter," in *Proc. 2017 Int. Conf. Unmanned Aircraft Systems (ICUAS)*, Miami, FL, USA, Jun. 13–16, 2017, pp. 1432–1437. https://doi.org/10.1109/ICUAS.2017.7991322.

ARTICLE IN PRESS

ARTICLE IN PRESS

96. Y. Zhong, W. Zhang, and Y. Zhang, "Sensor fault diagnosis for unmanned quadrotor helicopter via adaptive two-stage extended Kalman filter," in *Proc. 2017 Int. Conf. Sensing, Diagnostics, Prognostics, and Control (SDPC)*, Shanghai, China, Aug. 16–18, 2017, pp. 493–498. https://doi.org/10.1109/SDPC.2017.99.

Author Contributions: Zheyi Xie: conceived and designed the study, conducted the data analysis, and drafted the initial manuscript. Yucai Long: contributed to data collection, interpretation of results, and critical revision of the manuscript for important intellectual content. Both authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Funding: Research on the Application of Digital Twin Technology in the Optimization of CNC Machining Paths (23C0943)

Conflicts of Interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

ARTICLE IN PRESS